Math 132

Differential Topology

Lecture Notes

Textbook:

Victor Guillemin & Alan Pollack, *Differential Topology* (1974)

Topics:

(i) Smooth manifolds and smooth maps

(ii) Transversality and intersection theory

(iii) Oriented intersection theory and applications

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Contents

Chapter 1

Smooth Manifolds

1.1 Smooth Manifolds

$\mathbb{R}^k$: k-dimensional Euclidean space, i.e. a point $x \in \mathbb{R}^k$ is a k-tuple $x = (x_1, \ldots, x_k)$ of real numbers.

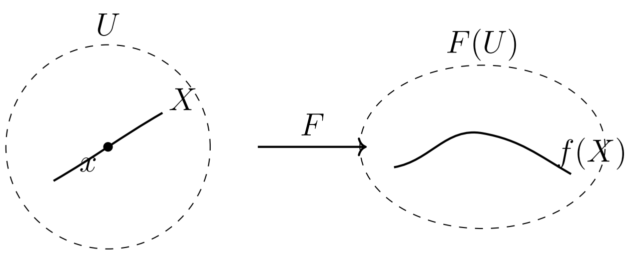
Definition 1.1. Smooth Map (Open Sets)

For open sets $U \subset \mathbb{R}^k$ and $V \subset \mathbb{R}^\ell$, a map $f\colon U \to V$ is called *smooth* (or C^∞) if all the partial derivatives $\frac{\partial^n f}{\partial x_{i_1} \cdots \partial x_{i_n}}$ exist and are continuous.

More generally, for arbitrary subsets $X \subset \mathbb{R}^k$ and $Y \subset \mathbb{R}^\ell$:

Definition 1.2. Smooth Map (Arbitrary Subsets)

A map $f\colon X \to Y$ is called *smooth* if for each $x \in X$ there exists an open $U \subset \mathbb{R}^k$ containing x and a smooth map $F\colon U \to \mathbb{R}^\ell$ such that $F = f$ on $U \cap X$.



Proposition 1.3. Composition of Smooth Maps

If $f\colon X \to Y$ and $g\colon Y \to Z$ are smooth, then $g \circ f\colon X \to Z$ is also smooth. (Easy exercise.)

Definition 1.4. Diffeomorphism

A map $f\colon X \to Y$ is called a *diffeomorphism* if it is bijective and both f and f^{-1} are smooth.

Remark 1.5. Meta-definition: Differential Topology

Differential topology is the study of properties of "spaces" which are invariant under diffeomorphism.

A particularly nice class of "spaces" is:

Definition 1.6. Smooth Manifold

A subset $M \subset \mathbb{R}^k$ is called a *smooth manifold of dimension m* (or simply a *smooth m-manifold*) if it is locally diffeomorphic to $\mathbb{R}^m$, i.e. if each $x \in M$ has a neighborhood $W \cap M$ that is diffeomorphic to an open $U \subset \mathbb{R}^m$.
A diffeomorphism $\phi\colon U \to W \cap M$ is called a *parametrization* of the region $W \cap M$.
The inverse diffeomorphism $\phi^{-1}\colon W \cap M \to U$ is called a *coordinate system* on $W \cap M$.

Example 1.7. The Unit Sphere

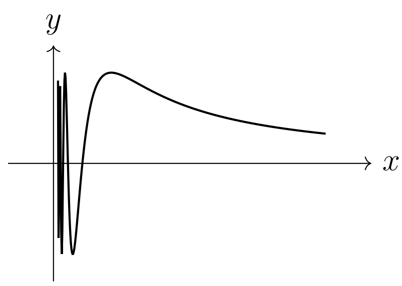
The unit sphere $S^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$ is a smooth 2-manifold (i.e. a smooth surface).
For instance, the diffeomorphism

$$(x, y) \mapsto (x, y, \sqrt{1 - x^2 - y^2}) \qquad (x^2 + y^2 < 1)$$

parametrizes the region $z > 0$ of S^2.
By interchanging the roles of x, y, z and changing the signs, we obtain parametrizations of regions which cover S^2.
More generally, $S^{n-1} = \{x \in \mathbb{R}^n \mid |x| = 1\}$ is a smooth $(n-1)$-manifold.

Example 1.8. Graph of $y = \sin(1/x)$

$\{(x, y) \in \mathbb{R}^2 \mid x \neq 0,\ y = \sin(1/x)\}$ is a smooth 1-manifold.



Remark 1.9. Extrinsic vs. Intrinsic Definition

Note that the definition of smooth manifolds we gave is "extrinsic" in that M is embedded in $\mathbb{R}^k$. We can also give a better, "intrinsic" definition as follows.

Definition 1.10. Topological Manifold

A Hausdorff, second countable topological space M is called a *topological m-manifold* if it is locally homeomorphic to $\mathbb{R}^m$.

Definition 1.11. Coordinate Chart

A *coordinate chart* is a pair (U, x) consisting of open $U \subset M$ and a homeomorphism $x \colon U \to \mathbb{R}^m$ onto its image.

Two charts (U, x) and (V, y) on M are *smoothly related* if the transition function $y \circ x^{-1} \colon x(U \cap V) \to y(U \cap V)$ is a diffeomorphism.

An *atlas* on M is a collection $\mathcal{A} = \{(U_\alpha, x_\alpha)\}_{\alpha \in A}$ of smoothly related charts which cover M (i.e. $\bigcup_{\alpha \in A} U_\alpha = M$).

A *smooth structure* on M is a maximal atlas. (Any atlas can be completed to a unique smooth structure.)

Definition 1.12. Smooth Manifold (Intrinsic)

A *smooth m-manifold* is a topological m-manifold equipped with a smooth structure.

Chapter 2

Derivatives and Tangents

2.1 Derivatives and Tangents

- For any smooth map $f\colon U \to V$ where $U \subset \mathbb{R}^k$ and $V \subset \mathbb{R}^\ell$ are open, the *derivative* at $x \in U$ is the linear map
$$df_x\colon \mathbb{R}^k \to \mathbb{R}^\ell$$
defined by the directional derivative
$$df_x(h) = \lim_{t\to 0} \frac{f(x+th) - f(x)}{t}, \qquad h \in \mathbb{R}^k.$$
It can be represented by the $\ell \times k$ Jacobian matrix $\left(\frac{\partial f_i}{\partial x_j}(x)\right)_{\substack{i=1,\dots,\ell \\ j=1,\dots,k}}$.
(After translation, this is the best approximation of f by a (nonhomogeneous) linear map, at x.)

- By chain rule, for any commutative triangle of smooth maps
$$\begin{tikzcd} U \arrow[r, "f"] \arrow[dr, "g\circ f"'] & V \arrow[d, "g"] \\ & W \end{tikzcd}$$
we have a commutative triangle of linear maps
$$\begin{tikzcd} \mathbb{R}^k \arrow[r, "df_x"] \arrow[dr, "d(g\circ f)_x"'] & \mathbb{R}^\ell \arrow[d, "dg_y"] \\ & \mathbb{R}^m \end{tikzcd}$$

Theorem 2.1. Dimension is a Diffeomorphism Invariant

If f is a diffeomorphism between open sets $U \subset \mathbb{R}^k$ and $V \subset \mathbb{R}^\ell$, then k must be equal to ℓ, and $df_x\colon \mathbb{R}^k \to \mathbb{R}^\ell$ must be nonsingular.

Proof. Since $f^{-1} \circ f = \mathrm{id}_U$, we have $d(f^{-1})_{f(x)} \circ df_x = \mathrm{id}_{\mathbb{R}^k}$. Similarly $df_x \circ d(f^{-1})_{f(x)} = \mathrm{id}_{\mathbb{R}^\ell}$. Thus df_x has a two-sided inverse, and it follows that $k = \ell$. $\square$

2.1.1 Tangent Spaces

Definition 2.2. Tangent Space

For a smooth m-manifold $M \subset \mathbb{R}^k$, choose a parametrization $g\colon U \to M \subset \mathbb{R}^k$ of a neighborhood $g(U)$ of x in M, with $g(u) = x$.
Set the *tangent space of* M *at* x to be

$$T_x M := dg_u(\mathbb{R}^m),$$

i.e. the image of $dg_u\colon \mathbb{R}^m \to \mathbb{R}^k$.

Proposition 2.3. Tangent Space is Well-Defined

This definition does not depend on the choice of parametrization.

Proof. If $h\colon V \to M \subset \mathbb{R}^k$ is another parametrization of a neighborhood $h(V)$ of x in M, with $h(v) = x$, then

$$\begin{tikzcd} & \mathbb{R}^k \\ U \arrow[ur, "g"] & V \arrow[u, "h"'] \arrow[l, "h^{-1}\circ g"'] \end{tikzcd}$$

where $U' = g^{-1}(g(U) \cap h(V))$ and $V' = h^{-1}(g(U) \cap h(V))$, so

$$\begin{tikzcd} & \mathbb{R}^k \\ \mathbb{R}^m \arrow[ur, "dg_u"] & \mathbb{R}^m \arrow[u, "dh_v"'] \arrow[l, "d(h^{-1}\circ g)_u"'] \end{tikzcd}$$

and thus $dg_u(\mathbb{R}^m) = dh_v(\mathbb{R}^m)$. $\square$

Proposition 2.4. Tangent Space is m-Dimensional

$T_x M$ is m-dimensional (i.e. $dg_u\colon \mathbb{R}^m \to \mathbb{R}^k$ is injective).

Proof. Since $g^{-1}\colon g(U) \to U$ is smooth, there is an open set W containing x and a smooth map $F\colon W \to \mathbb{R}^m$ that coincides with g^{-1} on $W \cap g(U)$. Then we have

$$\begin{tikzcd} & W \arrow[d, "F"] \\ U \arrow[ur, "g"] \arrow[r, hook] & \mathbb{R}^m \end{tikzcd}$$

where $U' = g^{-1}(W \cap g(U))$, so

$$\begin{tikzcd} & \mathbb{R}^k \arrow[d, "dF_x"] \\ \mathbb{R}^m \arrow[ur, "dg_u"] \arrow[r, equal] & \mathbb{R}^m \end{tikzcd}$$

and thus dg_u is injective. $\square$

2.1.2 Derivative of Maps Between Manifolds

Definition 2.5. Derivative of a Map Between Manifolds

For a smooth map $f\colon M \to N$ between smooth manifolds $M \subset \mathbb{R}^k$ and $N \subset \mathbb{R}^\ell$, the *derivative*

$$df_x\colon T_xM \to T_{f(x)}N$$

is defined by $df_x(v) = dF_x(v)$, $v \in T_xM$, for any smooth map $F\colon W \to \mathbb{R}^\ell$ that coincides with f on $W \cap M$.

Proposition 2.6. Derivative is Well-Defined

$dF_x(v) \in T_{f(x)}N$ and it does not depend on the particular choice of F.

Proof. Choose parametrizations $g\colon U \to M \subset \mathbb{R}^k$, $h\colon V \to N \subset \mathbb{R}^\ell$ for neighborhoods $g(U)$ of x in M and $h(V)$ of $f(x)$ in N. Replacing U by a smaller set if necessary, we may assume that $g(U) \subset W$ and that $f \circ g(U) \subset h(V)$.
Then we can consider the commutative diagram

$$\begin{tikzcd} W \arrow[r, "F"] & \mathbb{R}^\ell \\ U \arrow[u, "g"] \arrow[r, "h^{-1}\circ f\circ g"'] & V \arrow[u, "h"'] \end{tikzcd}$$

of smooth maps between open sets. Taking derivatives,

$$\begin{tikzcd} \mathbb{R}^k \arrow[r, "dF_x"] & \mathbb{R}^\ell \\ \mathbb{R}^m \arrow[u, "dg_u"] \arrow[r, "d(h^{-1}\circ f\circ g)_u"'] & \mathbb{R}^n \arrow[u, "dh_v"'] \end{tikzcd}$$

and it is clear that $dF_x(T_xM) \subset T_{f(x)}N$.
Moreover, $df_x = dh_v \circ d(h^{-1} \circ f \circ g)_u \circ (dg_u)^{-1}$, so it is independent of F. Therefore, $df_x\colon T_xM \to T_{f(x)}N$ is well-defined. $\square$

Remark 2.7. Chain Rule for Manifolds

As before, chain rule still holds: $M \xrightarrow{f} N \xrightarrow{g} P$ implies $T_xM \xrightarrow{df_x} T_{f(x)}N \xrightarrow{dg_{f(x)}} T_{g(f(x))}P$, with $d(g \circ f)_x = dg_{f(x)} \circ df_x$.
And for any diffeomorphism $f\colon M \to N$, $df_x\colon T_xM \to T_{f(x)}N$ is an isomorphism.

Chapter 3

Inverse Function Theorem and Immersions

3.1 Inverse Function Theorem and Immersions

Suppose M, N are smooth manifolds of the same dimension and $f\colon M \to N$ is a smooth map.

Definition 3.1. Local Diffeomorphism

We say f is a *local diffeomorphism* at $p \in M$ if f maps any sufficiently small open neighborhood U of p diffeomorphically onto an open set $f(U)$.

Clearly (from last lecture), if f is a local diffeomorphism at $p \in M$, then $df_p\colon T_pM \to T_{f(p)}N$ is an isomorphism.

The converse is also true!

Theorem 3.2. Inverse Function Theorem

If $f\colon M \to N$ is a smooth map whose derivative df_p at $p \in M$ is an isomorphism, then f is a local diffeomorphism at p.

Proof. Easily follows from the Inverse Function Theorem for open subsets of Euclidean spaces, using local parametrizations. □

Morally: local behavior of smooth maps is completely determined by their derivatives.

3.1.1 Immersions

More generally, when $\dim M \leq \dim N$, a map $f\colon M \to N$ is said to be an *immersion* at $p \in M$ if df_p is injective. If it is an immersion at every point, we simply call it an *immersion*.

Example 3.3. Immersions

1. $S^1 \to \mathbb{R}^2$: a figure-eight curve $\bigcirc \to 8$.
2. $\mathbb{R} \to T^2$ with irrational slope. (The image may not be a manifold.)

Theorem 3.4. Local Immersion Theorem

If $f\colon M \to N$ is an immersion at $p \in M$, then there exist local coordinates around $p \in M$ and $f(p) \in N$ such that

$$f(x_1, \dots, x_m) = (x_1, \dots, x_m, 0, \dots, 0).$$

Proof. Choose any local parametrizations $\phi\colon U \to M$, $U \subset \mathbb{R}^m$, with $\phi(0) = p$, and $\psi\colon V \to N$, $V \subset \mathbb{R}^n$, with $\psi(0) = f(p)$.
By making U smaller if necessary, we may assume $f \circ \phi(U) \subset \psi(V)$ to get a commutative diagram

$$\begin{tikzcd} M \arrow[r, "f"] & N \\ U \arrow[u, "\phi"] \arrow[r, "g"'] & V \arrow[u, "\psi"'] \end{tikzcd}$$

where $g = \psi^{-1} \circ f \circ \phi$.
By the assumption, $dg_0\colon \mathbb{R}^m \to \mathbb{R}^n$ is injective. Hence, by change of basis in $\mathbb{R}^n$, we may assume

$$dg_0 = \begin{pmatrix} I_m \\ 0 \end{pmatrix}$$

where I_m is the $m \times m$ identity matrix and 0 is the $(n-m) \times m$ zero matrix.
Now define $G\colon U \times \mathbb{R}^{n-m} \to \mathbb{R}^n$ by

$$(x, z) \mapsto g(x) + (0, z).$$

Then $dG_0 = \begin{pmatrix} I_m & \\ & I_{n-m} \end{pmatrix} = I_n$. Therefore, by the Inverse Function Theorem, G is a local diffeomorphism at 0.
Note also that $g = G \circ \iota$, where $\iota\colon U \to U \times \mathbb{R}^{n-m}$ is the canonical immersion $x \mapsto (x, 0)$.
It follows that, after shrinking U and V sufficiently, we have

$$\begin{tikzcd} M \arrow[r, "f"] & N \\ U \arrow[u, "\phi"] \arrow[r, "\iota"'] & V \arrow[u, "\psi \circ G"'] \end{tikzcd}$$

as desired. ∎

Corollary 3.5. Immersion is an Open Condition

If f is an immersion at p, then it is an immersion in a neighborhood of p. That is, *immersion is an open condition.*

3.1.2 Embeddings

Definition 3.6. Embedding

An immersion $f\colon M \to N$ that is injective and *proper* (preimage of every compact set is compact; automatic when M is compact) is called an *embedding*.

Theorem 3.7. Embeddings Map onto Submanifolds

An embedding $f\colon M \to N$ maps M diffeomorphically onto a submanifold $f(M)$ of N.

Proof. It suffices to show that the image of any open subset $W \subset M$ is an open subset of $f(M)$.
If $f(W)$ is not open in $f(M)$, there must be a sequence of points $q_i \in f(M) \setminus f(W)$ that converge to a point $q \in f(W)$.
By injectivity, each q_i has exactly one preimage $p_i \in M \setminus W$, and q has preimage $p \in W$.
By properness, $\{p, p_i \mid i \in \mathbb{N}\}$ is compact, and hence $\{p_i\}$ must have a subsequence converging to some $p' \in \{p, p_i \mid i \in \mathbb{N}\} \subset M$.
Since $f(p') = \lim f(p_{i_j}) = \lim q_{i_j} = q = f(p)$, p' must be p.
However, this contradicts our assumption that W is open.
Therefore, f is an open map, and the result follows from the local immersion theorem. $\square$

Chapter 4

Submersions

4.1 Submersions

Definition 4.1. Submersion

$f\colon M \to N$ is called a *submersion* at $p \in M$ if df_p is surjective. If it is a submersion at every point, it is simply called a *submersion*.

Theorem 4.2. Local Submersion Theorem

If $f\colon M \to N$ is a submersion at $p \in M$, then there exist local coordinates around $p \in M$ and $f(p) \in N$ such that

$$f(x_1, \dots, x_m) = (x_1, \dots, x_n).$$

Proof. Just as in the proof of the local immersion theorem, we start from local parametrizations

$$\begin{tikzcd} M \arrow[r, "f"] & N \\ U \arrow[u, "\phi"] \arrow[r, "g"'] & V \arrow[u, "\psi"'] \end{tikzcd}$$

where $U \subset \mathbb{R}^m$, $V \subset \mathbb{R}^n$, $\phi(0) = p$, $\psi(0) = f(p)$, and $g = \psi^{-1} \circ f \circ \phi$.
By the assumption, $dg_0\colon \mathbb{R}^m \to \mathbb{R}^n$ is surjective, so by change of basis in $\mathbb{R}^m$, we may assume

$$dg_0 = \begin{pmatrix} I_n & 0 \end{pmatrix}$$

where I_n is the $n \times n$ identity and 0 is $n \times (m - n)$.
Now define $G\colon U \to \mathbb{R}^m$ by

$$x \mapsto (g(x), x_{n+1}, \dots, x_m).$$

Then $dG_0 = I_m$, and G is a local diffeomorphism at 0.
Note also that $g = \pi \circ G$, where $\pi\colon \mathbb{R}^m \to \mathbb{R}^n$, $(x_1, \dots, x_m) \mapsto (x_1, \dots, x_n)$, is the canonical submersion.

It follows that, after shrinking U and V sufficiently, we have

$$\begin{tikzcd} M \arrow[r, "f"] & N \\ U \arrow[u, "\phi\circ G^{-1}"] \arrow[r, "\pi"'] & V \arrow[u, "\psi"'] \end{tikzcd}$$

as desired. □

Corollary 4.3. Submersion is an Open Condition

If f is a submersion at p, then it is a submersion in a neighborhood of p. That is, *submersion is an open condition.*

4.1.1 Regular and Critical Values

Definition 4.4. Regular Value

For a smooth map $f: M \to N$ of manifolds, a point $q \in N$ is called a *regular value* for f if $df_p: T_pM \to T_qN$ is surjective at every point $p \in f^{-1}(q)$.

Theorem 4.5. Preimage Theorem

If q is a regular value of $f: M \to N$, then the preimage $f^{-1}(q)$ is a submanifold of M with dimension

$$\dim f^{-1}(q) = \dim M - \dim N.$$

Proof. For any $p \in f^{-1}(q)$, thanks to the local submersion theorem, we can choose local coordinates around p and q so that

$$f(x_1, \ldots, x_m) = (x_1, \ldots, x_n).$$

Thus, near p, $f^{-1}(q)$ is just the set of points $(0, \ldots, 0, x_{n+1}, \ldots, x_m)$, and the functions $x_{n+1}, \ldots, x_m$ form a local coordinate system on a neighborhood of $p \in f^{-1}(q)$. □

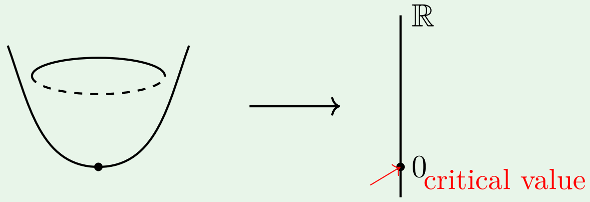
Definition 4.6. Critical Value

A point $q \in N$ that is not a regular value is called a *critical value.*

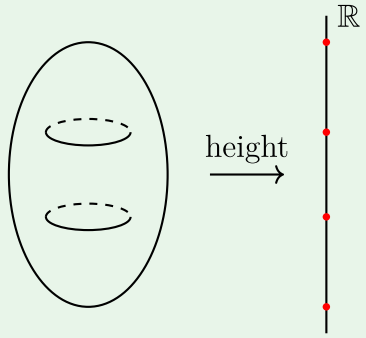
Later, we'll learn that the set of critical values is measure zero (Sard's theorem), so almost every point in N is a regular value of f.

Example 4.7. Regular and Critical Values

1. $\mathbb{R}^2 \to \mathbb{R}$, $(x, y) \mapsto x^2 + y^2$. The only critical value is 0.



2. $T^2 \to \mathbb{R}$ (height function on the torus). There are critical values at the top, bottom, and the two saddle points.



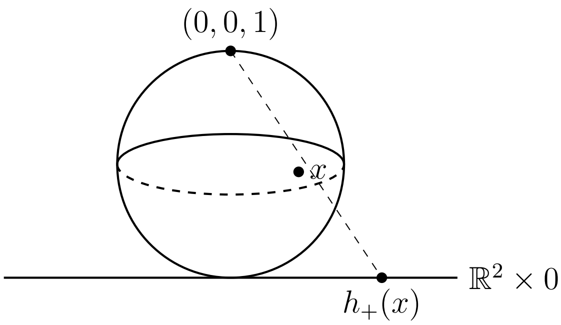
4.1.2 Application: Fundamental Theorem of Algebra

As an application of these notions, let's prove the *fundamental theorem of algebra*: every non-constant complex polynomial $P(z)$ must have a zero.

First, observe that if M is compact, $q \in N$ is a regular value, and $\dim M = \dim N$, then $f^{-1}(q)$ is a finite set. Moreover, $\# f^{-1}(q)$ is a locally constant function of q (as q ranges only through regular values). (If $p_1, \dots, p_k$ are points of $f^{-1}(q)$, then we can choose pairwise disjoint neighborhoods $U_1, \dots, U_k$ mapping diffeomorphically onto $V_1, \dots, V_k$. We may then take $V = V_1 \cap \cdots \cap V_k - f(M - U_1 - \cdots - U_k)$.)

Proof of the fundamental theorem of algebra. Consider the unit sphere $S^2 \subset \mathbb{R}^3$ and the stereographic projection

$$h_+ \colon S^2 \setminus \{(0,0,1)\} \to \mathbb{R}^2 \times 0 \subset \mathbb{R}^3.$$



We identify $\mathbb{R}^2 \times 0$ with $\mathbb{C}$. Say $P(z) = a_0 z^n + a_1 z^{n-1} + \cdots + a_n$ with $a_0 \neq 0$. The polynomial map $P \colon \mathbb{C} \to \mathbb{C}$ corresponds to the map

$$\begin{aligned} f \colon S^2 &\to S^2 \\ x &\mapsto h_+^{-1} \circ P \circ h_+(x) \quad \text{for } x \neq (0,0,1), \\ (0,0,1) &\mapsto (0,0,1). \end{aligned}$$

The resulting map is smooth, even in a neighborhood of $(0,0,1)$. (Let h_- be the stereographic projection from the south pole $(0,0,-1)$. Then $h_+ h_-^{-1}(z) = \frac{z}{|z|^2} = \frac{1}{\bar{z}}$, and $h_- \circ f \circ h_-^{-1}(z) = \frac{z^n}{\bar{a}_0 + \bar{a}_1 z + \cdots + \bar{a}_n z^n}$ is smooth at 0.)

Moreover, f has only finitely many critical points, since $P'(z)$ is not identically zero. Hence, the set of regular values of f is connected, and thus $\#f^{-1}(q)$ must be constant on this set. Since $\#f^{-1}(q)$ can't be zero everywhere, it is zero nowhere. Thus f is surjective, and P must have a zero. $\square$

Chapter 5

Transversality, Homotopy, and Stability

5.1 Transversality

Definition 5.1. Transversality of Submanifolds

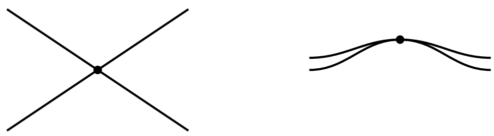
We say two smooth submanifolds $M, M' \subset N$ are *transversal* at $p \in M \cap M'$ if $T_pM + T_pM' = T_pN$. If they are transversal at every point $p \in M \cap M'$, we simply say they are *transversal*, and write $M \pitchfork M'$.

Definition 5.2. Transversality of a Map

More generally, we say a smooth map $f\colon M \to N$ is *transversal* to a submanifold $M' \subset N$ (abbreviated $f \pitchfork M'$) if

$$\operatorname{im}(df_p) + T_qM' = T_qN \quad \text{for every } p \in f^{-1}(M'),$$

where $q = f(p)$.



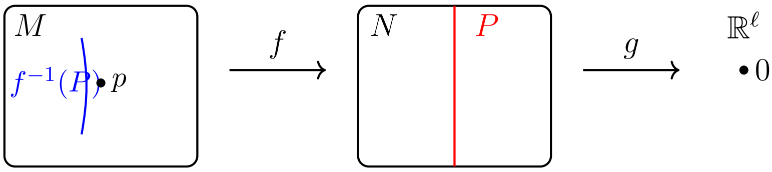
transversal in $\mathbb{R}^2$ not transversal in $\mathbb{R}^2$

The following theorem generalizes the preimage theorem for regular values:

Theorem 5.3. Transversal Preimage Theorem

If $f\colon M \to N$ is a smooth map transversal to a submanifold $P \subset N$, then the preimage $f^{-1}(P) \subset M$ is a submanifold of M, whose codimension (i.e. $\dim M - \dim f^{-1}(P)$) equals that of $P \subset N$ (i.e. $\dim N - \dim P$).

Proof. We need to show that every point $p \in f^{-1}(P)$ has a neighborhood U in M such that $f^{-1}(P) \cap U$ is a manifold.
Let $q = f(p)$. In a neighborhood of q, we may write P as the zero set of a collection of independent functions $g_1, \ldots, g_\ell$, where $\ell = \operatorname{codim} P = \dim N - \dim P$.



Then, near p, $f^{-1}(P)$ is the zero set of $(g_1 \circ f, \dots, g_\ell \circ f) = g \circ f$, so it's enough to show that 0 is a regular value of $g \circ f$.
Since $d(g \circ f)_p = dg_q \circ df_p$ and $dg_q \colon T_q N \to \mathbb{R}^\ell$ is surjective with $\ker dg_q = T_q P \subset T_q N$, 0 is a regular value of $g \circ f$ if and only if $\operatorname{im}(df_p) + T_q P = T_q N$, which is exactly our transversality assumption. □

Corollary 5.4. Intersection of Transversal Submanifolds

If $M, M' \subset N$ are two transversal submanifolds, then $M \cap M'$ is also a submanifold, with

$$\operatorname{codim}(M \cap M') = \operatorname{codim} M + \operatorname{codim} M'.$$

Remark 5.5. Additivity of Codimension

Additivity of codimension is very natural. Think: cutting out a submanifold locally using $k = \operatorname{codim} M$ and $\ell = \operatorname{codim} M'$ independent functions.

5.2 Homotopy and Stability

Definition 5.6. Homotopy

We say two smooth maps $f_0, f_1 \colon M \to N$ are *homotopic*, abbreviated $f_0 \sim f_1$, if there exists a smooth map $F \colon M \times I \to N$ such that $F(x, 0) = f_0(x)$ and $F(x, 1) = f_1(x)$. We call such F a *homotopy* between f_0 and f_1.
(Here $f_t(x) = F(x, t)$ is a smoothly evolving family of maps.)

Homotopy is an equivalence relation on smooth maps from M to N, and the equivalence class to which a map belongs is called its *homotopy class*.

Definition 5.7. Stability

A property of smooth maps is called *stable* if whenever $f_0 \colon M \to N$ possesses the property and $f_t \colon M \to N$ is a homotopy of f_0, then, for some $\varepsilon > 0$, each f_t with $t < \varepsilon$ also possesses the property.

Theorem 5.8. Stability Theorem

The following classes of smooth maps of a *compact* manifold M into a manifold N are stable classes:

- local diffeomorphisms,
- immersions,
- submersions,

- maps transversal to a given closed submanifold $P \subset N$,
- embeddings,
- diffeomorphisms.

(See Guillemin–Pollack pp. 36–37 for the proofs.)

Chapter 6

Sard's Theorem

6.1 Sard's Theorem

Theorem 6.1. Sard's Theorem

Let $f\colon M \to N$ be a smooth map, and let

$$C = \{x \in M \mid \operatorname{rank} df_x < \dim N\}$$

be the set of critical points. Then, the set of critical values, $f(C)$, has measure zero in N.

Proof. Thanks to second-countability of manifolds, it suffices to prove it for $f\colon U \to \mathbb{R}^p$, $U \overset{\text{open}}{\subset} \mathbb{R}^n$.
We proceed by induction on n. Note, it is trivial for $n = 0$.
Let

$$C_1 = \{x \in U \mid df_x = 0\}$$

and more generally

$$C_k = \left\{ x \in U \;\middle|\; \frac{\partial^j f}{\partial x_{s_1} \cdots \partial x_{s_j}}(x) = 0 \text{ for all } j \leq k \text{ and } 1 \leq s_1, \ldots, s_j \leq n \right\}.$$

Thus we have a descending sequence of closed sets

$$C \supset C_1 \supset C_2 \supset \cdots.$$

We will prove the theorem in three steps:
Step 1: $f(C - C_1)$ has measure zero.
Step 2: $f(C_k - C_{k+1})$ has measure zero, for $k \geq 1$.
Step 3: $f(C_k)$ has measure zero for k sufficiently large.

Proof of Step 1. We may assume $p \geq 2$, since $C = C_1$ when $p = 1$.
We will use:

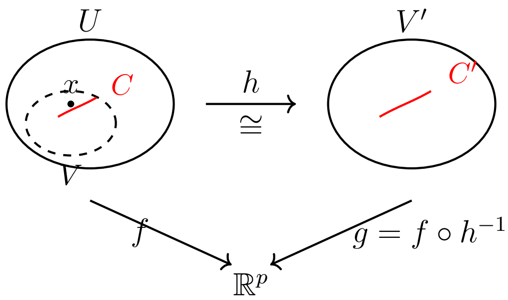
Theorem 6.2. Fubini

A measurable set $A \subset \mathbb{R}^p = \mathbb{R} \times \mathbb{R}^{p-1}$ must have measure zero if it intersects each hyperplane (const.) $\times \mathbb{R}^{p-1}$ in a set of $(p-1)$-dimensional measure zero.

For each $x \in C - C_1$, we will find an open neighborhood $V \subset \mathbb{R}^n$ so that $f(V \cap C)$ has measure zero.
Since $x \notin C_1$, some partial derivative, say $\frac{\partial f_1}{\partial x_1}$, is nonzero at x. Consider the map $h \colon U \to \mathbb{R}^n$,

$$x \mapsto (f_1(x), x_2, \ldots, x_n).$$

Since dh_x is nonsingular, it is a local diffeomorphism; it maps some neighborhood V of x diffeomorphically on to an open set $V' \subset \mathbb{R}^n$.



Set $g := f \circ h^{-1}$. Then the set C' of critical points of g is $h(V \cap C)$, and the set $g(C')$ of critical values of g is $f(V \cap C)$.
Note, g carries hyperplanes into hyperplanes; let

$$g^t \colon (t \times \mathbb{R}^{n-1}) \cap V' \to t \times \mathbb{R}^{p-1}$$

denote the restriction of g. Moreover, since

$$\left(\frac{\partial g_i}{\partial x_j}\right) = \begin{pmatrix} 1 & 0 \\ * & \frac{\partial g_i^t}{\partial x_j} \end{pmatrix},$$

a point of $t \times \mathbb{R}^{n-1}$ is critical for g^t iff it is critical for g.
By induction hypothesis, the set of critical values of g^t has measure zero in $t \times \mathbb{R}^{p-1}$.
Hence, by Fubini's theorem, the set

$$g(C') = f(V \cap C)$$

has measure zero, which completes Step 1. □

Proof of Step 2. (Similar to Step 1.)
For each $x \in C_k - C_{k+1}$, some $(k+1)$-st derivative is nonzero, so let's say $w(x) = \frac{\partial^k f_r}{\partial x_{s_1} \cdots \partial x_{s_k}}$ vanishes at x but $\frac{\partial w}{\partial x_1}$ does not.
Then the map $h\colon U \to \mathbb{R}^n$, $x \mapsto (w(x), x_2, \dots, x_n)$, carries some neighborhood V diffeomorphically onto an open set $V' \subset \mathbb{R}^n$.
Note, h carries $C_k \cap V$ into the hyperplane $0 \times \mathbb{R}^{n-1}$. Hence, we can consider $g := f \circ h^{-1}\colon V' \to \mathbb{R}^p$ and its restriction $\bar{g}\colon (0 \times \mathbb{R}^{n-1}) \cap V' \to \mathbb{R}^p$.
By induction, the set of critical values of $\bar{g}$ has measure 0 in $\mathbb{R}^p$, and in particular, $\bar{g} \circ h(C_k \cap V) = f(C_k \cap V)$ has measure zero. □

Proof of Step 3. Let $I^n \subset U$ be a cube with edge δ. We will prove that if k is sufficiently large, $f(C_k \cap I^n)$ has measure zero. (The constant depends only on f and I^n.)
By Taylor's theorem, $f(x+h) = f(x) + R(x,h)$ where $\|R(x,h)\| \le c\|h\|^{k+1}$ for $x \in C_k \cap I^n$, $x + h \in I^n$.
Now, subdivide I^n into r^n cubes of edge $\frac{\delta}{r}$.
Then, for each cube I' in the subdivision containing a point $x \in C_k$, any other point of I' can be written as $x + h$ with $\|h\| \le \sqrt{n}\,\frac{\delta}{r}$, so

$$\mathrm{Volume}(f(C_k \cap I')) \le \left(2c\left(\sqrt{n}\,\frac{\delta}{r}\right)^{k+1}\right)^p.$$

Therefore,

$$\mathrm{Volume}(f(C_k \cap I^n)) \le r^n \cdot \left(2c\left(\sqrt{n}\,\frac{\delta}{r}\right)^{k+1}\right)^p = c' \cdot r^{n-(k+1)p}$$

which holds for any $r \in \mathbb{Z}_{>0}$, so we can send $r \to \infty$. And hence if $k+1 > \frac{n}{p}$, $f(C_k \cap I^n)$ must have measure zero.
This completes the proof of Sard's theorem. □

□

Chapter 7

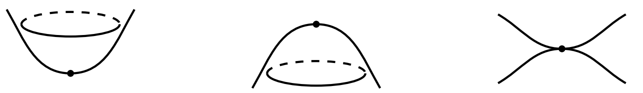
Morse Functions

7.1 Morse Functions

Consider a smooth function $f\colon M \to \mathbb{R}$. If $x \in M$ is a regular point of f (i.e. if $df_x \neq 0$), we know that we can choose a local coordinate around x so that f is simply the first coordinate function.

What can we say at critical points?

Recall from calculus that a critical point x of $f\colon \mathbb{R}^k \to \mathbb{R}$ is called *nondegenerate* if the *Hessian matrix* $H = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)$ is nonsingular at x.



$\lambda_1, \lambda_2 > 0$ (index 0) $\quad$ $\lambda_1, \lambda_2 < 0$ (index 2) $\quad$ $\lambda_1 > 0 > \lambda_2$ (index 1)

Proposition 7.1. Nondegenerate Critical Points are Isolated

Nondegenerate critical points are isolated from the other critical points.

Proof. Consider the map $\nabla f\colon \mathbb{R}^k \to \mathbb{R}^k$, $x \mapsto \left(\frac{\partial f}{\partial x_1}(x), \ldots, \frac{\partial f}{\partial x_k}(x)\right)$. Then $\nabla f(x) = 0$, since x is a critical point. Nondegeneracy means $d(\nabla f)_x$ is a local diffeomorphism at x. Thus f has no other critical point in a small neighborhood of x. $\square$

7.1.1 Nondegeneracy on Manifolds

Nondegeneracy of critical points makes sense on manifolds, thanks to the following:

Lemma 7.2. Nondegeneracy is Independent of Coordinates

Suppose $f\colon \mathbb{R}^k \to \mathbb{R}$ has a nondegenerate critical point at 0 and ψ is a diffeomorphism with $\psi(0) = 0$. Then $f \circ \psi$ also has a nondegenerate critical point at 0.

Proof. Let $f' = f \circ \psi$. Then

$$\frac{\partial f'}{\partial x_i}(x) = \sum_\alpha \frac{\partial f}{\partial x_\alpha}(\psi(x)) \frac{\partial \psi_\alpha}{\partial x_i}(x),$$

and

$$\frac{\partial^2 f'}{\partial x_i \partial x_j}(0) = \sum_\alpha \sum_\beta \frac{\partial^2 f}{\partial x_\alpha \partial x_\beta}(0) \frac{\partial \psi_\alpha}{\partial x_i}(0) \frac{\partial \psi_\beta}{\partial x_j}(0) + \sum_\alpha \frac{\partial f}{\partial x_\alpha}(0) \frac{\partial^2 \psi_\alpha}{\partial x_i \partial x_j}(0).$$

The last sum vanishes since 0 is a critical point. That is, the new Hessian is $H' = (d\psi_0)^t H(d\psi_0)$. Since H and $d\psi_0$ are nonsingular, so is H'. □

Definition 7.3. Morse Function

A function $f: M \to \mathbb{R}$ whose critical points are all nondegenerate is called a *Morse function*.

Morse functions are extremely important! They tell us a great deal about the topology of M, and much of modern topology is based on this idea. It'll take another whole course to talk about Morse theory, but let me just mention:

Morse lemma: If $a \in M$ is a nondegenerate critical point of f, then there is a local coordinate system $(x_1, \dots, x_m)$ around a such that

$$f = f(a) + \sum h_{ij} x_i x_j \quad \text{near } a,$$

i.e. f is locally a quadratic polynomial.

7.1.2 Abundance of Morse Functions

As an application of Sard's theorem, let's show that the vast majority of functions are Morse functions.

Theorem 7.4. Generic Functions are Morse

Let $f: M \to \mathbb{R}$ be a function, and suppose our manifold M sits inside $\mathbb{R}^N$. For each $a = (a_1, \dots, a_N) \in \mathbb{R}^N$, define

$$f_a := f + a_1 x_1 + \cdots + a_N x_N,$$

where $x_1, \dots, x_N$ are the usual coordinate functions on $\mathbb{R}^N$. Then, for almost every $a \in \mathbb{R}^N$, f_a is a Morse function.

Proof. Let's first prove the result in $\mathbb{R}^k$:

Lemma 7.5. Local Morse Genericity

Let $f: U \subset \mathbb{R}^k \to \mathbb{R}$ be a smooth function. Then, for almost all $a = (a_1, \dots, a_k) \in \mathbb{R}^k$,

$$f_a := f + a_1 x_1 + \cdots + a_k x_k$$

is a Morse function on U.

Proof. Note that $(df_a)_p = \left(\frac{\partial f_a}{\partial x_1}(p), \dots, \frac{\partial f_a}{\partial x_k}(p)\right) = \nabla f(p) + a$, so p is a critical point of f_a iff $\nabla f(p) = -a$. Moreover, it's nondegenerate iff $d(\nabla f)_p$ is nonsingular, i.e. if p is a regular point of ∇f.

Thanks to Sard, $-a$ is a regular value of ∇f for almost all $a \in \mathbb{R}^k$. □

Now let's turn this into a global result. Suppose $p \in M$ and $x_1, \dots, x_N$ are the standard

coordinate functions on $\mathbb{R}^N$. Then, some m of these coordinate functions, $x_{i_1}, \ldots, x_{i_m}$, constitute a coordinate system in a neighborhood of p.
Hence, we can cover M with countably many such open subsets U_α, and it suffices to show that f_a is Morse on U_α for almost every $a \in \mathbb{R}^N$.
For convenience, assume that $(x_1, \ldots, x_m)$ is a coordinate system on U_α. Let $S_\alpha = \{a \in \mathbb{R}^N \mid f_a \text{ is not Morse on } U_\alpha\}$.
Then, by the lemma, $S_\alpha \cap (\mathbb{R}^m \times \{c\})$ has measure zero in $\mathbb{R}^m$, for any $c = (c_{m+1}, \ldots, c_N)$. By Fubini, it follows that S_α has measure zero. $\square$

Chapter 8

Whitney Embedding Theorem

8.1 Whitney Embedding Theorem

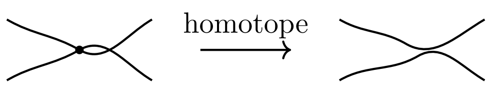
So far, our m-manifold M has been embedded in $\mathbb{R}^k$ where k may be enormous. Can we embed it in a smaller Euclidean space?

Theorem 8.1. Strong Whitney Embedding Theorem

Any smooth m-manifold (with $m > 0$) can be embedded in $\mathbb{R}^{2m}$.

In this lecture, we'll only prove a weaker version where we embed it in $\mathbb{R}^{2m+1}$; going further down to $\mathbb{R}^{2m}$ requires a method known as the "Whitney trick."

Intuitively, it's clear why it should be possible to embed M^m into $\mathbb{R}^{2m+1}$: since $m+m < 2m+1$, there's enough room to wiggle things around.



in $\mathbb{R}^2$ in $\mathbb{R}^3$

Remark 8.2

Whitney's result is the best possible linear bound. In fact, when m is a power of 2, there's no embedding $\mathbb{RP}^m \hookrightarrow \mathbb{R}^{2m-1}$.

Now, let's prove:

Theorem 8.3. Injective Immersion in $\mathbb{R}^{2m+1}$

Every m-manifold admits an injective immersion in $\mathbb{R}^{2m+1}$.

Proof. If $M^m \subset \mathbb{R}^k$, we will produce a linear projection $\mathbb{R}^k \to \mathbb{R}^{2m+1}$ that restricts to an injective immersion of M. Proceeding inductively, it suffices to show that: if $f\colon M \to \mathbb{R}^\ell$ is an injective immersion with $\ell > 2m+1$, then there exists a unit vector $a \in \mathbb{R}^\ell$ such that the composition of f with the projection map $\pi_a\colon \mathbb{R}^\ell \to H_a := \{b \in \mathbb{R}^\ell \mid b \perp a\}$ is still an injective immersion.
Define maps

$$\begin{aligned} h\colon M \times M \times \mathbb{R} &\to \mathbb{R}^\ell, & (x, y, t) &\mapsto t(f(x) - f(y)), \\ g\colon TM &\to \mathbb{R}^\ell, & (x, v) &\mapsto df_x(v). \end{aligned}$$

Since $\ell > 2m+1$, by Sard, there exists a point $a \in \mathbb{R}^\ell \setminus \{0\}$ belonging to neither image. Then $\pi_a \circ f\colon M \to H_a$ is injective, since $\pi_a \circ f(x) = \pi_a \circ f(y) = ta$ implies either $t = 0$ (which means $x = y$ by injectivity of f) or $h(x, y, 1/t) = a$ (contradiction to our choice of a).
Moreover, $\pi_a \circ f$ is an immersion, since $d(\pi_a \circ f)_x(v) = 0$ implies $\pi_a \circ df_x(v) = 0$, so $df_x(v) = ta$ for some $t \neq 0$ (by immersivity of f), which gives $g(x, v/t) = a$, a contradiction. □

For compact manifolds, injective immersions are the same as embeddings, so we have just proved the (weaker version of) embedding theorem in the compact case.

For non-compact manifolds, we need to modify the immersion to make it proper, which is a topological, not a differential problem. The fundamental technique for such generalization is the *partition of unity*.

Definition 8.4. Partition of Unity

Let M be a smooth manifold. A *partition of unity* $\{\rho_i\}_{i\in I}$ is a set of smooth functions $\rho_i\colon M \to \mathbb{R}$ such that

(a) $\rho_i \geq 0$,

(b) each $x \in M$ has a neighborhood on which all but finitely many ρ_i are identically zero,

(c) $\sum_{i\in I} \rho_i(x) = 1$ for all $x \in M$.

If $\{U_\alpha\}_{\alpha\in A}$ is an open cover of M, then $\{\rho_i\}_{i\in I}$ is *subordinate* to $\{U_\alpha\}_{\alpha\in A}$ if each $\operatorname{supp}\rho_i = \overline{\rho_i^{-1}(\mathbb{R}\setminus\{0\})}$ is contained in some U_α.

Theorem 8.5. Existence of Partitions of Unity

Every open cover $\{U_\alpha\}_{\alpha\in A}$ admits a countable partition of unity $\{\rho_i\}_{i\in I}$ subordinate to it.

(For now, let's take this theorem for granted.)

Corollary 8.6. Existence of Proper Functions

On any manifold M, there exists a proper function $f\colon M \to \mathbb{R}$.

Proof. Let $\{U_\alpha\}_{\alpha\in A}$ be the set of open subsets of M with compact closure. Choose a partition of unity $\{\rho_i\}_{i\in\mathbb{Z}_{>0}}$ subordinate to it. Define $f = \sum_{i=1}^\infty i\,\rho_i$. Then $f^{-1}([-j,j]) \subset \bigcup_{i=1}^j \operatorname{supp}\rho_i$, so f is proper. □

Now we're ready to prove the:

Theorem 8.7. Whitney Embedding Theorem (Weak)

Every m-manifold embeds in $\mathbb{R}^{2m+1}$.

Proof. Begin with an injective immersion $M \to \mathbb{R}^{2m+1}$. Composing with a diffeomorphism of $\mathbb{R}^{2m+1}$ into its unit ball, $z \mapsto \frac{z}{1+|z|^2}$, we obtain an injective immersion $f\colon M \to \mathbb{R}^{2m+1}$ with $|f(x)| < 1$ for all $x \in M$.

Let $g\colon M \to \mathbb{R}$ be a proper function, and define a new injective immersion

$$F\colon M \to \mathbb{R}^{2m+2}, \qquad x \mapsto (f(x), g(x)).$$

Recall that $\pi_a \circ F\colon M \to H_a \cong \mathbb{R}^{2m+1}$ is an injective immersion for almost every $a \in S^{2m+1}$, so pick an a that is neither of the sphere's two poles.
We claim that $\pi_a \circ F$ is proper. In fact, we will show that, for any c, there exists d such that $(\pi_a \circ F)^{-1}(\bar{B}_c) \subset g^{-1}(\bar{B}_d)$, which is compact since g is proper.
If the claim is false, there should be a sequence of points $\{x_i\}$ in M for which $|\pi_a \circ F(x_i)| < c$ but $g(x_i) \to \infty$.
By the definition of π_a,

$$w_i := \frac{1}{g(x_i)}\Big(F(x_i) - \pi_a \circ F(x_i)\Big) \in \mathbb{R}^{2m+2}$$

is a multiple of a. As $i \to \infty$,

$$w_i = \frac{F(x_i)}{g(x_i)} - \frac{\pi_a \circ F(x_i)}{g(x_i)} = \left(\frac{f(x_i)}{g(x_i)}, 1\right) - \frac{\pi_a \circ F(x_i)}{g(x_i)} \to (0, \dots, 0, 1).$$

Since each w_i is a multiple of a, so should be the limit. This contradicts our choice of a, and therefore $\pi_a \circ F$ is proper. $\square$

Chapter 9

Manifolds with Boundary

9.1 Manifolds with Boundary

Consider the closed *half-space*

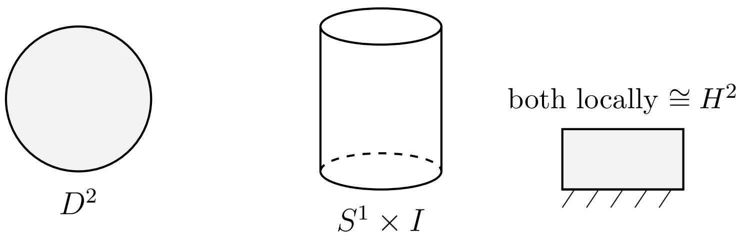
$$H^m := \{(x_1, \dots, x_m) \in \mathbb{R}^m \mid x_m \geq 0\}.$$

The boundary ∂H^m is defined to be the hyperplane $\mathbb{R}^{m-1} \times 0 \subset \mathbb{R}^m$.

Definition 9.1. Manifold with Boundary

A subset $M \subset \mathbb{R}^k$ is called a *smooth m-manifold with boundary* if it is locally diffeomorphic to H^m.

The *boundary* ∂M is the set of all points of M which correspond to points of ∂H^m under such a local parametrization. Its complement is called the *interior* $\mathrm{Int}(M) = M - \partial M$.

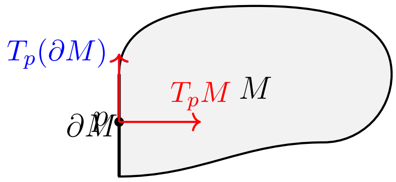


Proposition 9.2. Boundary is a Manifold

If M is a smooth m-manifold with boundary, ∂M is a well-defined smooth $(m-1)$-manifold (without boundary).

Proof. For any open neighborhood U of $x \in \partial H^m$ in H^m, $U \cap \partial H^m$ is an open subset of $\partial H^m \cong \mathbb{R}^{m-1}$, so it suffices to show that any diffeomorphism between open subsets of H^m sends boundary points to boundary points.
Suppose $f\colon U \to V$ is such a diffeomorphism, and that it maps some $x \in \partial U$ to an interior point $y = f(x)$ of V. But then f^{-1} must map an open neighborhood of y in $\mathbb{R}^m$ (contained in V) onto an open neighborhood of x in $\mathbb{R}^m$, which contradicts $x \in \partial U$.
Therefore, ∂M is a well-defined smooth $(m-1)$-manifold. $\square$

The *tangent space* T_pM is defined just as before, so that it is a full m-dimensional vector space, even if $p \in \partial M$. In case $p \in \partial M$, there is a natural inclusion $T_p(\partial M) \hookrightarrow T_pM$ (($m-1$)-dimensional into m-dimensional).



9.1.1 Generating Examples

Here's one method for generating examples:

Lemma 9.3. Preimage of a Regular Value

Let M be a manifold (without boundary) and let $g\colon M \to \mathbb{R}$ have 0 as a regular value. Then $\{x \in M \mid g(x) \geq 0\}$ is a smooth manifold with boundary $g^{-1}(0)$.

Proof. The set where $g > 0$ is open in M and hence a submanifold of the same dimension. At points of $g^{-1}(0)$, g is locally equivalent to the canonical submersion, and the lemma is obvious. □

Example 9.4. Unit Disk

The unit disk $D^m = \{x \in \mathbb{R}^m \mid 1 - \sum_{i=1}^m x_i^2 \geq 0\}$ is a smooth manifold with boundary $\partial D^m = S^{m-1}$.

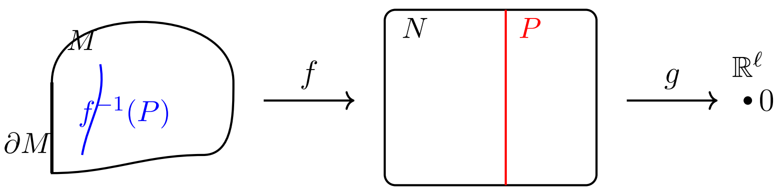
9.1.2 Preimage Theorem with Boundary

We also have a version of the preimage theorem:

Theorem 9.5. Preimage Theorem for Manifolds with Boundary

Let $f\colon M \to N$ be a smooth map from a manifold M with boundary to a manifold N (without boundary), and suppose that both $f\colon M \to N$ and $\partial f := f|_{\partial M}\colon \partial M \to N$ are transversal with respect to a submanifold $P \subset N$.
Then, $f^{-1}(P) \subset M$ is a manifold with boundary $\partial f^{-1}(P) = f^{-1}(P) \cap \partial M$, and $\operatorname{codim}(f^{-1}(P) \subset M) = \operatorname{codim}(P \subset N)$.

Proof. We only need to examine $f^{-1}(P)$ in a neighborhood of a point $x \in f^{-1}(P) \cap \partial M$, thanks to the earlier preimage theorem.
As usual, in some neighborhood of $y = f(x) \in P$, there's a submersion g to $\mathbb{R}^\ell$, $\ell = \operatorname{codim}(P \subset N)$, such that $P = g^{-1}(0)$ in this neighborhood.



Pre-composing with a local parametrization $\phi\colon U \subset H^m \to M$ around x, we obtain a map $h = g \circ f \circ \phi\colon U \subset H^m \to \mathbb{R}^\ell$ which is a submersion. By definition of smoothness, h

extends to a smooth map $\tilde{h}$ defined on a neighborhood $\tilde{U}$ of u in $\mathbb{R}^m$, and the preimage $\tilde{h}^{-1}(0) \subset \tilde{U}$ is a manifold.

Now, the transversality assumption on ∂f implies that the last coordinate x_m is regular on $\tilde{h}^{-1}(0)$ at u. Therefore, by the previous lemma, $\tilde{h}^{-1}(0) \cap \{x_m \geq 0\}$ is a manifold with boundary $\tilde{h}^{-1}(0) \cap \{x_m = 0\}$. $\square$

Chapter 10

1-Manifolds and Some Consequences

10.1 1-Manifolds and Some Consequences

Theorem 10.1. Classification of 1-Manifolds

Every compact, connected 1-manifold with boundary is diffeomorphic to S^1 or $I = [0, 1]$.

(See the Appendix of Milnor or Guillemin–Pollack for the proof. The idea is to parametrize the 1-manifold by arc length, in a maximal way.)

Corollary 10.2. Boundary Has Even Cardinality

The boundary of any compact 1-manifold with boundary consists of an even number of points.

As we will see, this seemingly trivial corollary has numerous nontrivial applications.

Later, in the next chapter, we will refine this statement by counting those boundary points with sign (orientation). Such signed count is always 0, and this fact is at the heart of *oriented intersection theory*.

10.1.1 No Retraction Lemma

Let M be a compact manifold with boundary.

Lemma 10.3. No Retraction

There's no smooth map $f\colon M \to \partial M$ that leaves ∂M pointwise fixed. That is, there is no retraction of M onto ∂M.

Proof. Suppose there were such a map f. Let $y \in \partial M$ be a regular value for f (which exists thanks to Sard). Since y is certainly a regular value for the identity map $f|_{\partial M}$ as well, $f^{-1}(y)$ is a smooth 1-manifold, with boundary

$$\partial f^{-1}(y) = f^{-1}(y) \cap \partial M = \{y\}.$$

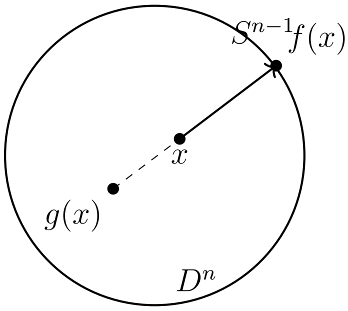
But $f^{-1}(y)$ is also compact, so that $\#|\partial f^{-1}(y)|$ must be even, a contradiction. □

10.1.2 Brouwer Fixed Point Theorem

Theorem 10.4. Smooth Brouwer Fixed Point Theorem

Any smooth map $g\colon D^n \to D^n$ has a fixed point.

Proof. Suppose g has no fixed point. Then we can construct a retraction $f\colon D^n \to S^{n-1}$ which is impossible from the lemma:



$$f(x) = x + tu, \quad \text{where } u = \frac{x - g(x)}{\|x - g(x)\|} \text{ and } t = -x \cdot u + \sqrt{1 - x \cdot x + (x \cdot u)^2}.$$

Note t is strictly positive. $\square$

Often, one can use an approximation theorem to pass to the continuous case, and this theorem is an example:

Theorem 10.5. Brouwer Fixed Point Theorem

Any continuous map $G\colon D^n \to D^n$ has a fixed point.

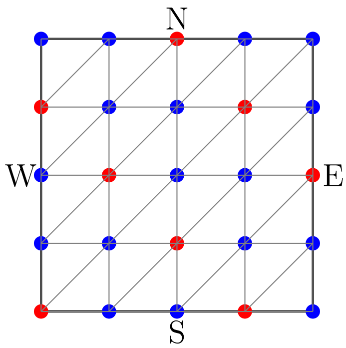
Proof. Given $\varepsilon > 0$, thanks to the Weierstrass approximation theorem, there is a polynomial function $P_0\colon \mathbb{R}^n \to \mathbb{R}^n$ with $\|P_0(x) - G(x)\| < \varepsilon$ for $x \in D^n$.
However, P_0 may send points of D^n into points outside of D^n, so to correct this we set $P(x) = \frac{P_0(x)}{1+\varepsilon}$.
Clearly, $P\colon D^n \to D^n$ and $\|P(x) - G(x)\| < \frac{2\varepsilon}{1+\varepsilon} < 2\varepsilon$ for $x \in D^n$.
Suppose G has no fixed points. Then the continuous function $\|G(x) - x\|$ must take on a minimum $\mu > 0$ on D^n. Choosing $P\colon D^n \to D^n$ as above, with $\|P(x) - G(x)\| < \mu$ for all $x \in D^n$, we have $P(x) \neq x$ for all $x \in D^n$.
Since P is smooth, this contradicts the Smooth Brouwer fixed point theorem. $\square$

10.1.3 The Hex Theorem

The following is just for fun:

Theorem 10.6. Hex Theorem

There's no draw in the game of Hex. That is, if we color the vertices of a triangulation of I^2 in 2 colors, say red and blue, then either the red connects E–W or the blue connects N–S.



Proof.
Suppose there is a coloring for which neither red nor blue connects the two opposite sides. Then, we can construct a continuous map $f\colon I^2 \to I^2$ without any fixed point, which contradicts the Brouwer fixed point theorem.
We construct such f as follows: For each vertex v of the triangulation, set

$$f(v) = \begin{cases} v + (\varepsilon, 0) & \text{if } v \text{ is colored in red and connected to W,} \\ v - (\varepsilon, 0) & \text{if } v \text{ is colored in red and not connected to W,} \\ v + (0, \varepsilon) & \text{if } v \text{ is colored in blue and connected to S,} \\ v - (0, \varepsilon) & \text{if } v \text{ is colored in blue and not connected to S.} \end{cases}$$

Extend linearly on each simplex. Then f has no fixed points. □

Chapter 11

Transversality

11.1 Transversality

Earlier, we saw that transversality is stable under small perturbations. Today, we'll use Sard's theorem to show that transversal maps are *generic* in the sense that any smooth map can be deformed by an arbitrarily small amount into a transversal map.

(Think as a smooth family of maps $f_s\colon M \to N$, $s \in S$.)

Theorem 11.1. Transversality Theorem

Suppose $F\colon M \times S \to N$ is a smooth map of manifolds, where only M has ∂, and let $P \subset N$ be a submanifold. If both F and $F|_{\partial M \times S}$ are transversal to P, then for almost every $s \in S$, both f_s and $f_s|_{\partial M}$ are transversal to P.

Proof. The preimage $Q = F^{-1}(P) \subset M \times S$ is a submanifold with boundary $\partial Q = Q \cap (\partial M \times S)$. Let $\pi\colon Q \to S$ be the projection.

We'll show that ① $f_s \pitchfork P$ whenever s is a regular value for π, and ② $f_s|_{\partial M} \pitchfork P$ whenever s is a regular value for $\pi|_{\partial Q}$. The theorem will then follow from Sard.

Note that the second claim is a special instance of the first claim, for the map $F|_{\partial M\times S}\colon \partial M \times S \to N$ of boundaryless manifolds, so it suffices to show ①.

In order to show $f_s \pitchfork P$, we need to show that, for any $x \in M$ with $f_s(x) = y \in P$, the composition $T_xM \xrightarrow{d(f_s)_x} T_yN \xrightarrow{p} T_yN/T_yP$ is surjective. (Here $d(f_s)_x(v) = dF_{(x,s)}(v, 0)$.)

Since $F \pitchfork P$, we know that

$$T_{(x,s)}(M \times S) = T_xM \times T_sS \xrightarrow{dF_{(x,s)}} T_yN \xrightarrow{p} T_yN/T_yP$$

is surjective. Moreover, by the regularity assumption, $d\pi_{(x,s)}\colon T_{(x,s)}Q \to T_sS$ is surjective.

Since $T_{(x,s)}Q \subset T_xM \times T_sS$ is in the kernel of $p \circ dF_{(x,s)}$, it follows that the restriction $p \circ dF_{(x,s)}|_{T_xM\times 0} = p \circ d(f_s)_x$ is surjective, as desired. $\square$

Corollary 11.2. Transversal Maps are Generic (Euclidean Target)

Let M be a smooth manifold with boundary, $f\colon M \to \mathbb{R}^\ell$ a smooth map, and $P \subset \mathbb{R}^\ell$ a submanifold. Then for almost every $s \in \mathbb{R}^\ell$, the translated map $f_s(x) := f(x) + s$

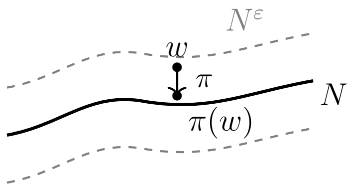
satisfies $f_s \pitchfork P$ and $f_s|_{\partial M} \pitchfork P$.
In particular, s may be chosen arbitrarily small.

Proof. Define $F\colon M \times \mathbb{R}^\ell \to \mathbb{R}^\ell$ by $F(x, s) = f(x) + s$. For any fixed $x \in M$, the map $s \mapsto F(x, s)$ is a translation, hence a submersion. So F itself is a submersion, as is $F|_{\partial M \times \mathbb{R}^\ell}$. In particular, both are transversal to any submanifold $P \subset \mathbb{R}^\ell$.
The Transversality Theorem then gives $f_s \pitchfork P$ and $f_s|_{\partial M} \pitchfork P$ for almost every $s \in \mathbb{R}^\ell$. Since the exceptional set has measure zero, it has empty interior, so s may be chosen arbitrarily small. $\square$

For an arbitrary target manifold $N \subset \mathbb{R}^\ell$, the proof of genericity of transversal maps is essentially the same; we just need to project the shifted map down to N.

Lemma 11.3. ε-Neighborhood Theorem

For any manifold $N \subset \mathbb{R}^\ell$, there is a smooth positive function $\varepsilon\colon N \to \mathbb{R}_{>0}$ such that each point $w \in N^\varepsilon := \{w \in \mathbb{R}^\ell \mid \|w - y\| < \varepsilon(y) \text{ for some } y \in N\}$ has a unique closest point $\pi(w)$ in N. Moreover, the map $\pi\colon N^\varepsilon \to N$ is a submersion.
(See pp. 71–72 of Guillemin–Pollack for the proof.)



Corollary 11.4. Submersive Family

Let $f\colon M \to N$ be a smooth map, where only M has boundary. Then there is an open ball S in some Euclidean space and a smooth map $F\colon M \times S \to N$ such that $F(x, 0) = f(x)$, and for any fixed $x \in M$, the map $S \to N$, $s \mapsto F(x, s)$, is a submersion. In particular, both F and $F|_{\partial M \times S}$ are submersions.

Proof. Let S be the unit open ball in $\mathbb{R}^\ell \supset N$, and define

$$F(x, s) = \pi\Big(f(x) + \varepsilon(f(x))s\Big).$$

Since $\pi\colon N^\varepsilon \to N$ restricts to the identity on N, $F(x, 0) = f(x)$. Since $s \mapsto f(x) + \varepsilon(f(x))s$ and π are compositions of two submersions, it is a submersion. $\square$

11.1.1 Transversality Homotopy and Extension Theorems

Theorem 11.5. Transversality Homotopy Theorem

For any smooth map $f\colon M \to N$ and a submanifold $P \subset N$, where only M has ∂, there exists a smooth map $g\colon M \to N$ homotopic to f such that $g \pitchfork P$ and $g|_{\partial M} \pitchfork P$.

Proof. For the family of maps F in the corollary, the transversality theorem implies that $f_s \pitchfork P$ and $f_s|_{\partial M} \pitchfork P$ for almost all $s \in S$, but each f_s is homotopic to f, with homotopy $M \times I \to N$, $(x, t) \mapsto F(x, ts)$. $\square$

We'll actually need a slightly stronger form of this theorem:

Theorem 11.6. Extension Theorem (Relative Transversality Homotopy)

In the setting as above, where P is a closed submanifold, suppose $C \subset M$ is a closed subset such that $f \pitchfork P$ on C and $f|_{\partial M} \pitchfork P$ on $C \cap \partial M$.
Then there exists a smooth map $g\colon M \to N$ homotopic to f such that $g \pitchfork P$, $g|_{\partial M} \pitchfork P$, and $g = f$ on a neighborhood of C.
(See pp. 72–73 of Guillemin–Pollack for the proof. The idea is to modify $F\colon M \times S \to N$ where $\tau\colon M \to [0,1]$ is a smooth bump function that is 0 on C and 1 outside of an open neighborhood, by setting $G\colon M \times S \to N$, $(x,s) \mapsto F(x, \tau(x)s)$.)

Corollary 11.7. Extension of Transversality

If $f\colon M \to N$ is a smooth map such that $f|_{\partial M}\colon \partial M \to N$ is transversal to $P \subset N$, then there exists a map $g\colon M \to N$ homotopic to f rel ∂ (i.e. $g|_{\partial M} = f|_{\partial M}$) such that $g \pitchfork P$.

Chapter 12

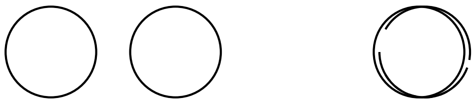
Mod 2 Intersection Theory

12.1 Mod 2 Intersection Theory

Consider two closed submanifolds $M_1, M_2 \subset N$ of complementary dimension (i.e. $\dim M_1 + \dim M_2 = \dim N$), where at least one of M_1 or M_2 is compact.

When they intersect transversely, $M_1 \cap M_2$ must be a compact 0-manifold (i.e. a finite set of points), so we can count the "intersection number" $\#(M_1 \cap M_2)$.

Even if their intersection is not transverse, we know we can homotope them to make it transverse.



$I_2(M_1, M_2) = 0$ $\qquad$ $I_2(M_1, M_2) = 1$

Depending on the choice of homotopy, the number of intersections might change, but we'll show that the *mod 2 intersection number* is well-defined.

Definition 12.1. Mod 2 Intersection Number

Suppose M_1 is a compact manifold, $f\colon M_1 \to N$ is a smooth map, $M_2 \subset N$ is a closed submanifold with $\dim M_1 + \dim M_2 = \dim N$. Choose any map g homotopic to f and transversal to M_2. Define the *mod 2 intersection number* of f with M_2, $I_2(f, M_2)$, to be

$$\#g^{-1}(M_2) \mod 2 \quad \in \mathbb{Z}/2\mathbb{Z}.$$

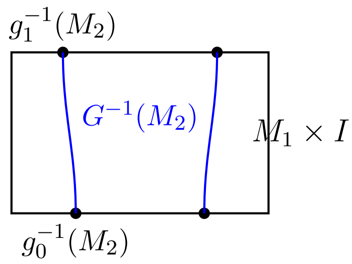
Theorem 12.2. Mod 2 Intersection Number is Well-Defined

If $g_0, g_1\colon M_1 \to N$ are homotopic and both transversal to $M_2 \subset N$, then $\#g_0^{-1}(M_2) = \#g_1^{-1}(M_2) \mod 2$.

In particular, the mod 2 intersection number is well-defined and invariant under homotopy.

Proof. Let $G\colon M_1 \times I \to N$ be a homotopy of g_0 and g_1. By the relative transversality homotopy theorem, we may assume $G \pitchfork M_2$. Then $G^{-1}(M_2)$ is a 1-dimensional submanifold of $M_1 \times I$ with boundary

$$\partial G^{-1}(M_2) = G^{-1}(M_2) \cap \partial(M_1 \times I) = g_0^{-1}(M_2) \times \{0\} \cup g_1^{-1}(M_2) \times \{1\}.$$



Since the boundary of any compact 1-manifold has an even number of points, $\#g_0^{-1}(M_2) + \#g_1^{-1}(M_2)$ is even. □

Example 12.3. Mod 2 Intersection Numbers

1. Let $D \subset \mathbb{R}^3$ be a disk and $S^1 \subset \mathbb{R}^3$ a circle, with ∂D unlinked from S^1. Then $I_2(D, S^1) = 0$, since D may be chosen disjoint from S^1. (Dim check: $\dim D + \dim S^1 = 2 + 1 = 3 = \dim \mathbb{R}^3$.)

2. Let $D \subset \mathbb{R}^3$ be a disk and $S^1 \subset \mathbb{R}^3$ a circle, with ∂D linked with S^1. Then $I_2(D, S^1) = 1$, since every disk bounded by ∂D must be punctured by S^1 an odd number of times.

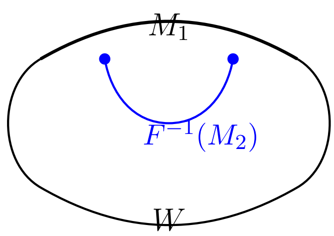
3. The core circle M of a Möbius band N: the mod 2 self-intersection number $I_2(M, M) = 1$.

12.1.1 Boundary Theorem

Theorem 12.4. Boundary Theorem

Suppose that M_1 is the boundary of some compact manifold W and $f\colon M_1 \to N$ is a smooth map. If f can be extended to all of W, then $I_2(f, M_2) = 0$ for any closed submanifold $M_2 \subset N$ of complementary dimension.

Proof. Let $F\colon W \to N$ extend f, which we may assume to be transversal to M_2, by the transversality homotopy theorem. Then $F^{-1}(M_2)$ is a compact 1-manifold with boundary $\partial F^{-1}(M_2) = f^{-1}(M_2)$, so $\#f^{-1}(M_2)$ must be even.



□

12.1.2 Mod 2 Degree

Definition 12.5. Mod 2 Degree

If $f\colon M \to N$ is a smooth map of a compact manifold M into a connected manifold N of the same dimension, define the *mod 2 degree* of f,

$$\deg_{\mathbb{Z}/2}(f), \text{ to be } I_2(f,\{y\}), \text{ for any point } y \in N.$$

Theorem 12.6. Mod 2 Degree is Well-Defined

Mod 2 degree is well-defined.

Proof. For any $y \in N$, we can homotope f to make it transversal to $\{y\}$. From Lecture 4, we know $y \mapsto I_2(f,\{y\})$ is locally constant. Since N is connected, it must be globally constant. □

Since $\deg_{\mathbb{Z}/2}$ is defined as an intersection number, we immediately obtain:

Theorem 12.7

The mod 2 degree is invariant under homotopy.

Theorem 12.8

If $M = \partial W$ and $f\colon M \to N$ extends to all of W, then $\deg_{\mathbb{Z}/2}(f) = 0$.

Example 12.9. Mod 2 Degree Examples

A constant map $c\colon M \to M$, $x \mapsto c$, has $\deg_{\mathbb{Z}/2}(c) = 0$.
The identity map $\mathrm{id}\colon M \to M$, $x \mapsto x$, has $\deg_{\mathbb{Z}/2}(\mathrm{id}) = 1$.
In particular, the identity map of a compact manifold (without boundary) is NOT homotopic to a constant map.
(In case $M = \partial W$, this implies that $\mathrm{id}\colon M \to M$ cannot be extended to all of W, a result we saw in Lecture 10.)

Chapter 13

Winding Number and Borsuk–Ulam Theorem

13.1 Winding Numbers

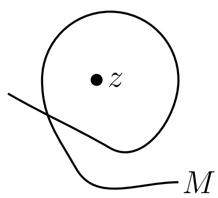
Definition 13.1. Winding Number

Let M be a compact, connected $(n-1)$-manifold and $f\colon M \to \mathbb{R}^n$ a smooth map. If $z \in \mathbb{R}^n$ is not in the image of f, consider the map

$$u\colon M \to S^{n-1}, \qquad x \mapsto \frac{f(x)-z}{|f(x)-z|}.$$

Define the *mod 2 winding number* of f around z to be the mod 2 degree of u, i.e.

$$\operatorname{wind}_{\mathbb{Z}/2}(f,z) := \deg_{\mathbb{Z}/2}(u).$$



Theorem 13.2. Winding Number Formula

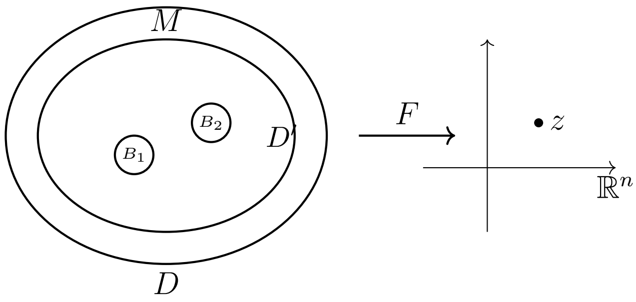
Suppose $M = \partial D$ for some compact n-manifold D with boundary, and let $F\colon D \to \mathbb{R}^n$ be a map extending f. If $z \in \mathbb{R}^n$ is a regular value of F that's not in the image of f, then

$$\operatorname{wind}_{\mathbb{Z}/2}(f,z) = \#F^{-1}(z) \mod 2.$$

Proof. If z is not in the image of F, then u extends to D, so $\deg_{\mathbb{Z}/2}(u) = 0$ in that case.
If $F^{-1}(z) = \{y_1, \dots, y_\ell\}$, then F is a local diffeomorphism at each y_i, $1 \le i \le \ell$, so we can choose a small ball B_i around y_i such that $u_i\colon \partial B_i \to S^{n-1}$ is bijective (and hence $\operatorname{wind}_{\mathbb{Z}/2}(f_i, z) = 1$), where $f_i := F|_{\partial B_i}$. We may assume that the balls are disjoint from each other and from $M = \partial D$.

But our previous observation (applied to $D' = D - \bigcup_{i=1}^{\ell} \mathrm{Int}(B_i)$) implies

$$\mathrm{wind}_{\mathbb{Z}/2}(f,z) = \mathrm{wind}_{\mathbb{Z}/2}(f_1,z) + \cdots + \mathrm{wind}_{\mathbb{Z}/2}(f_\ell,z) \mod 2 = \#F^{-1}(z) \mod 2.$$



$\square$

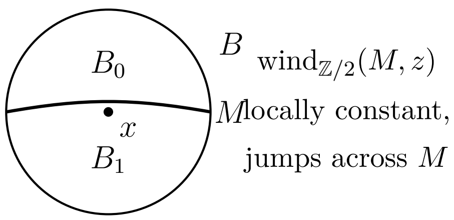
13.1.1 Jordan–Brouwer Separation Theorem

When M is a compact, connected hypersurface in $\mathbb{R}^n$ (i.e. codimension 1 submanifold), we can use the winding numbers to prove:

Theorem 13.3. Jordan–Brouwer Separation Theorem

The complement of a compact, connected hypersurface M in $\mathbb{R}^n$ consists of two connected open sets, the "outside" D_0 and the "inside" D_1. Moreover, $\bar{D}_1$ is a compact n-manifold with boundary $\partial \bar{D}_1 = M$.

Proof sketch. Let $D_a := \{z \in \mathbb{R}^n \setminus M \mid \mathrm{wind}_{\mathbb{Z}/2}(M,z) = a\}$ for each $a \in \mathbb{Z}/2$, so that $\mathbb{R}^n \setminus M = D_0 \sqcup D_1$. Since $\mathrm{wind}_{\mathbb{Z}/2}(M,z)$ is locally constant, they are both open.
For each $x \in M$, if B is a small open ball around x in $\mathbb{R}^n$, then $B \setminus M = B_0 \sqcup B_1$, where B_0, B_1 are two connected half balls, and $B_a = D_a \cap B$.



Connectivity of M implies D_a is connected, and compactness of M implies $\bar{D}_1$ is compact. $\square$

13.2 Borsuk–Ulam Theorem

Theorem 13.4. Borsuk–Ulam Theorem

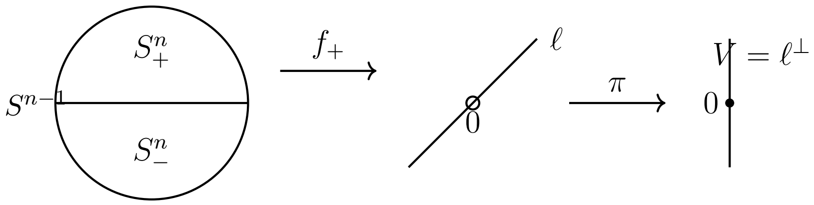
Let $f\colon S^n \to \mathbb{R}^{n+1} \setminus \{0\}$ be a smooth map such that $f(-x) = -f(x)$ for all $x \in S^n$. Then $\mathrm{wind}_{\mathbb{Z}/2}(f,0) = 1$.

Proof. Let's proceed by induction on n.
For $n = 1$, if $f\colon S^1 \to S^1$ is an antipodal map, then its lift $\tilde{f}\colon \mathbb{R} \to \mathbb{R}$ must satisfy $\tilde{f}(\theta + \pi) = \tilde{f}(\theta) + k\pi$ for some odd k, and $\deg_{\mathbb{Z}/2} f = k = 1 \pmod 2$.
Now assume the theorem for $n-1$, and let $f\colon S^n \to \mathbb{R}^{n+1} \setminus \{0\}$ be antipodal. The

idea is to compute $\mathrm{wind}_{\mathbb{Z}/2}(f, 0)$ by $I_2(f, R)$ for some ray R from 0.
Let $g = f|_{S^{n-1}}$ (the equator $S^n \cap \{x_{n+1} = 0\}$) and use Sard to choose a unit vector v that is a regular value for both $\frac{g}{|g|} \colon S^{n-1} \to S^n$ and $\frac{f}{|f|} \colon S^n \to S^n$.
Let $\ell = \mathbb{R} \cdot v$ be the corresponding line. Then $f \pitchfork \ell$, and g never intersects ℓ.
Now,

$$\mathrm{wind}_{\mathbb{Z}/2}(f, 0) = \deg_{\mathbb{Z}/2}\left(\frac{f}{|f|}\right) = \#\left(\frac{f}{|f|}\right)^{-1}(v) \mod 2 = \frac{1}{2}\#f^{-1}(\ell) = \#f_+^{-1}(\ell) \mod 2,$$

where $f_+ = f|_{S^n_+}$ is the restriction to the upper hemisphere.



By induction hypothesis,

$$\mathrm{wind}_{\mathbb{Z}/2}(f, 0) = \#f_+^{-1}(\ell) = \#(\pi \circ f_+)^{-1}(0) = \mathrm{wind}_{\mathbb{Z}/2}(\pi \circ g, 0) = 1 \mod 2.$$

$\square$

Corollary 13.5

If $f \colon S^n \to \mathbb{R}^{n+1} \setminus \{0\}$ is antipodal, then f intersects every line through 0 at least once.

Corollary 13.6

For any n smooth functions $g_1, \ldots, g_n$ on S^n, there exists a point $p \in S^n$ at which $g_i(p) = g_i(-p)$ for all $i = 1, \ldots, n$.

Proof. If not, then $f(x) = (g_1(x) - g_1(-x), \ldots, g_n(x) - g_n(-x), 0)$ is a smooth antipodal map $S^n \to \mathbb{R}^{n+1} \setminus \{0\}$ that misses the x_{n+1}-axis, contradicting the previous corollary applied with ℓ the x_{n+1}-axis. $\square$

Chapter 14

Orientation

14.1 Orientation

Definition 14.1. Equivalently Oriented Bases

Suppose V is a finite dimensional real vector space, and $\beta = \{v_1, \ldots, v_k\}$ and $\beta' = \{v_1', \ldots, v_k'\}$ are two ordered bases. Then there is a unique linear isomorphism $A\colon V \to V$ such that $\beta' = A\beta$. We say β and β' are *equivalently oriented* if $\det A > 0$.

This gives an equivalence relation on the set of all ordered bases of V, and there are exactly two equivalence classes ($\mathrm{GL}(V)$ has 2 components).

Definition 14.2. Orientation of a Vector Space

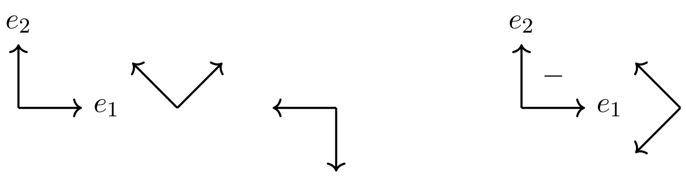
An *orientation* of V is a choice of one of those two equivalence classes.

An orientation induces a sign ($+$ or $-$) on any ordered basis of V.
($*$ When $V = 0$, we separately define its orientation to be a choice of a sign $\in \{+, -\}$.)

Example 14.3. Orientation of $\mathbb{R}^2$

Say, $V = \mathbb{R}^2$ and choose $[(e_1, e_2)]$ for the orientation. Then counterclockwise bases are positively oriented, while clockwise bases are negatively oriented.

positively oriented: negatively oriented:



Note, if V and W are oriented real vector spaces of the same dimension, then any isomorphism $A\colon V \to W$ either *preserves* or *reverses* orientation.

14.1.1 Orientation of Manifolds

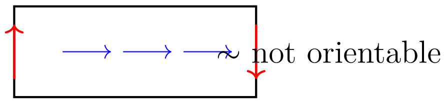
Definition 14.4. Orientation of a Manifold

An *orientation* of M, a manifold with boundary, is a smooth choice of orientations for all tangent spaces T_pM.
That is, around each $p \in M$, there is a local parametrization $h\colon U \to M$, $U \subset H^m \subset \mathbb{R}^m$, such that $dh_u\colon T_uU \cong \mathbb{R}^m \to T_{h(u)}M$ preserves orientation for all $u \in U$, where $\mathbb{R}^m$ is given the standard orientation $[(e_1, \ldots, e_m)]$.

A smooth map between oriented manifolds of the same dimension whose derivative preserves orientations at every point is called an *orientation-preserving map*.

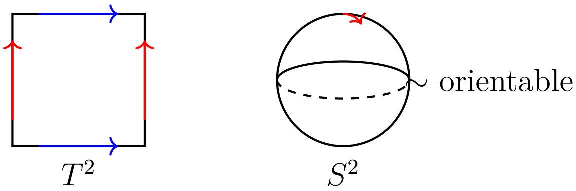
Example 14.5. Non-Orientable Manifolds

Not all manifolds can be oriented! The Möbius band is not orientable.



Example 14.6. Orientable Manifolds

T^2 (torus) and S^2 (sphere) are orientable.



Proposition 14.7. Two Orientations

A connected, orientable manifold with boundary admits exactly two orientations.

Proof. The set of points at which two orientations agree and the set where they disagree are both open. Consequently, two orientations of a connected manifold are either identical or opposite. □

14.1.2 Product and Boundary Orientations

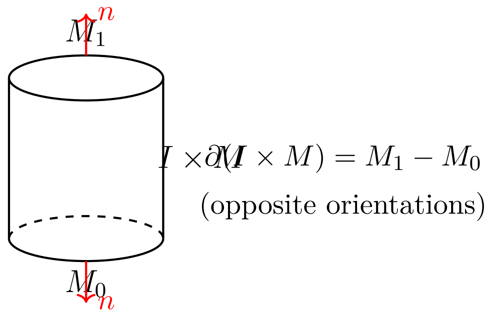
Remark 14.8. Product and Boundary Orientations

- The product $M \times N$ of two oriented manifolds (where one of them is boundaryless) acquires a *product orientation*, for if α and β are ordered bases for T_pM and T_qN, then $(\alpha \times 0, 0 \times \beta)$ is an ordered basis for $T_{(p,q)}(M \times N) \cong T_pM \times T_qN$.

- An orientation of M induces a *boundary orientation* on ∂M; this can be done by declaring the sign of any ordered basis $\beta = \{v_1, \ldots, v_{m-1}\}$ of $T_p(\partial M)$ to be the sign of the ordered basis $\{n_p, v_1, \ldots, v_{m-1}\}$, where n_p is the outward unit normal at p.

Example 14.9. Boundary Orientation of $I \times M$

For the cylinder $I \times M$, the boundary orientation gives $\partial(I \times M) = M_1 - M_0$, where M_0 and M_1 have opposite orientations.



14.1.3 Orienting Preimages

If $V = V_1 \oplus V_2$, the orientations on any two of V, V_1, V_2 induce a *direct sum orientation* on the third. (The ordering of summands is important!)

We can use this to orient $f^{-1}(P) =: Q$ if $f\colon M \to N$, $f \pitchfork P$, $f|_{\partial M} \pitchfork P$, and M, N, P are all oriented.

Let $N_x(Q; M)$ be the orthogonal complement of T_xQ in T_xM, so that

$$T_xM = N_x(Q; M) \oplus T_xQ.$$

By transversality,

$$T_yN = df_x(N_x(Q; M)) \oplus T_yP,$$

which gives an orientation on T_xQ!

Proposition 14.10. Boundary of Preimage

$\partial f^{-1}(P) = (-1)^{\operatorname{codim} P}(f|_{\partial M})^{-1}(P)$.

Proof sketch. $T_xM = N_x(Q; M) \oplus \mathbb{R} \cdot n_x \oplus T_x(\partial Q) = \mathbb{R} \cdot n_x \oplus N_x(Q; M) \oplus T_x((f|_{\partial M})^{-1}(P))$, so the orientations differ by $(-1)^{\operatorname{codim} Q} = (-1)^{\operatorname{codim} P}$. □

Chapter 15

Oriented Intersection Number

15.1 Oriented Intersection Number

Consider oriented manifolds M, N, P, with M compact, $P \subset N$ closed, and

$$\dim M + \dim P = \dim N.$$

If $f\colon M \to N$ is transversal to P, then $f^{-1}(P)$ is a finite number of points, each with an orientation number $\in \{\pm 1\}$.

Definition 15.1. Intersection Number

The *intersection number* $I(f, P) \in \mathbb{Z}$ is defined to be the sum of these orientation numbers.

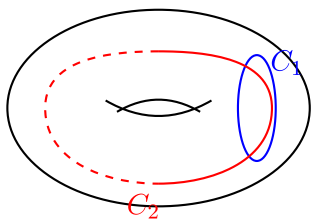
The orientation number at $x \in f^{-1}(P)$ is $+1$ (resp. -1) if the isomorphism

$$df_x(T_x M) \oplus T_{f(x)} P = T_{f(x)} N$$

is orientation preserving (resp. reversing). Note, the ordering is important!

Example 15.2. Circles on a Torus

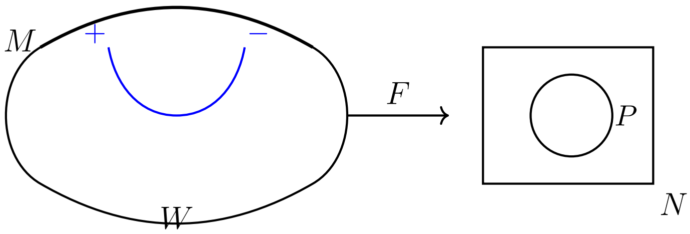
For the two standard circles C_1, C_2 on a torus, $I(C_1, C_2) = 1$ and $I(C_2, C_1) = -1$.



Proposition 15.3. Vanishing on Boundaries

If $M = \partial W$ for some compact, oriented W and $f\colon M \to N$ extends to W, then $I(f, P) = 0$ for any closed $P \subset N$ of complementary dimension.

Proof. The proof is the same as the analogous theorem in mod 2 intersection theory. Here, $F^{-1}(P)$ is a compact oriented 1-manifold, so the signed count of $\partial F^{-1}(P)$ is 0. $\square$



Proposition 15.4. Homotopy Invariance

Homotopic maps have the same intersection numbers.

Proof. The proof is analogous to the mod 2 case. If $F\colon I \times M \to N$ is a homotopy between $f_0, f_1\colon M \to N$, then

$$0 = I(\partial F, P) = I(f_1, P) - I(f_0, P),$$

since $\partial(I \times M) = M_1 - M_0$, so $\partial F^{-1}(P) = f_1^{-1}(P) - f_0^{-1}(P)$. □

15.1.1 Degree

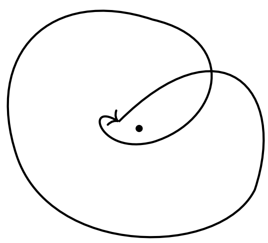
In the same manner, when N is connected and $\dim N = \dim M$, we define the *degree* of $f\colon M \to N$ to be

$$\deg(f) = I(f, \{y\}),$$

for any (positively oriented) point $y \in N$.

Example 15.5. Winding Number

The winding number of a closed curve in the plane around a point is a degree.



Example 15.6. Degree of $z \mapsto z^m$

The map $S^1 \to S^1$, $z \mapsto z^m$, has degree $m \in \mathbb{Z}$. In particular, z^m and $z^{m'}$ (as maps $S^1 \to S^1$) are not homotopic whenever $m \neq m'$.

15.1.2 Reformulation via the Diagonal

The following reformulation is useful.

Let M_1, M_2 be compact with $\dim M_1 + \dim M_2 = \dim N$. For any $f_1\colon M_1 \to N$ and $f_2\colon M_2 \to N$, we say they are *transversal*, $f_1 \pitchfork f_2$, if

$$(df_1)_{x_1}(T_{x_1}M_1) \oplus (df_2)_{x_2}(T_{x_2}M_2) = T_yN \tag{$*$}$$

whenever $f_1(x_1) = y = f_2(x_2)$.

Basic linear algebra shows $f_1 \pitchfork f_2 \iff f_1 \times f_2 \pitchfork \Delta$, where $f_1 \times f_2 \colon M_1 \times M_2 \to N \times N$ and $\Delta \subset N \times N$ is the *diagonal*.

Define $I(f_1, f_2)$ to be the sum of local intersection numbers, each of which is given by $+1$ (resp. -1) if the isomorphism $(*)$ is orientation preserving (resp. reversing).

Proposition 15.7. Intersection via Diagonal

$$I(f_1, f_2) = (-1)^{\dim(M_2)} I(f_1 \times f_2, \Delta).$$

Proof. This is linear algebra again. Let $U = (df_1)_{x_1}(T_{x_1}M_1)$ and $V = (df_2)_{x_2}(T_{x_2}M_2)$, and let $\{u_1, \dots, u_k\}$ and $\{v_1, \dots, v_\ell\}$ be their positively oriented ordered bases. We may assume $\{u_1, \dots, u_k, v_1, \dots, v_\ell\}$ is positively oriented for T_yN. Then, for $U \times V$,

$$\begin{aligned}
&\operatorname{sign}\{(u_1,0),\dots,(u_k,0),(0,v_1),\dots,(0,v_\ell),(u_1,u_1),\dots,(u_k,u_k),(v_1,v_1),\dots,(v_\ell,v_\ell)\}\\
&\qquad= \operatorname{sign}\{(u_1,0),\dots,(u_k,0),(0,v_1),\dots,(0,v_\ell),(0,u_1),\dots,(0,u_k),(v_1,0),\dots,(v_\ell,0)\}\\
&\qquad= (-1)^{\ell(\ell+k)+\ell k} \operatorname{sign}\{(u_1,0),\dots,(v_\ell,0),(0,u_1),\dots,(0,v_\ell)\}\\
&\qquad= (-1)^{\ell}.
\end{aligned}$$

□

Proposition 15.8. Homotopy Invariance of Pairs

If f_0 and g_0 are homotopic to f_1 and g_1, respectively, then $I(f_0, g_0) = I(f_1, g_1)$.

Proof. $f_t \times g_t$ is a homotopy between $f_0 \times g_0$ and $f_1 \times g_1$. □

In particular, if $M_1, M_2 \subset N$ are compact submanifolds of complementary dimension, then $I(M_1, M_2)$ is invariant under homotopy of either M_1 or M_2.

Note, from the definition,

$$I(M_1, M_2) = (-1)^{(\dim M_1)(\dim M_2)} I(M_2, M_1).$$

15.1.3 Euler Characteristic

Definition 15.9. Euler Characteristic

If M is a compact, orientable manifold, its *Euler characteristic* $\chi(M)$ is defined to be $I(\Delta, \Delta)$, where $\Delta \subset M \times M$ is the diagonal.

We immediately have:

Proposition 15.10. Odd-Dimensional Euler Characteristic

The self-intersection number of any odd-dimensional compact oriented manifold is 0. In particular, $\chi(M) = 0$ for any odd-dimensional compact oriented manifold.

Recall that $I_{\mathbb{Z}/2}(C, C) = 1$ for the central circle C in the Möbius strip. This shows that the Möbius strip is *not orientable* (for if it were, then $I_{\mathbb{Z}/2}(C, C) = I(C, C) \mod 2 = 0$).

Chapter 16

Lefschetz Fixed Point Theorem

16.1 Lefschetz Fixed Point Theorem

Let M be a compact, oriented manifold, and $f: M \to M$ a smooth map.

Definition 16.1. Global Lefschetz Number

The *global Lefschetz number* of f, denoted $L(f)$, is

$$L(f) = I(\Delta, \operatorname{graph}(f)).$$

The following are immediate consequences of the intersection theory. Note that fixed points of f correspond exactly to intersection points $\Delta \cap \operatorname{graph}(f)$.

Proposition 16.2. Homotopy Invariance of L

$L(f)$ is a homotopy invariant.

Theorem 16.3. Smooth Lefschetz Fixed Point Theorem

If $L(f) \neq 0$, then f has a fixed point.

Proposition 16.4. Lefschetz Number of Identity

If f is homotopic to the identity, then $L(f) = \chi(M)$.

In particular, if M admits a smooth map $f: M \to M$ homotopic to the identity that has no fixed points, then $\chi(M) = 0$.

Example 16.5. Euler Characteristic of S^1

Rotation of S^1 by an angle $\theta \neq 0 \mod 2\pi$ has no fixed points, so $\chi(S^1) = 0$.

16.1.1 Lefschetz Maps and Local Lefschetz Numbers

Definition 16.6. Lefschetz Map

We say a map $f: M \to M$ is *Lefschetz* if $\Delta \pitchfork \operatorname{graph}(f)$.

Proposition 16.7. Homotopy to Lefschetz

Every map $f\colon M \to M$ is homotopic to a Lefschetz map.

Note, $\Delta \pitchfork \operatorname{graph}(f)$ at (x,x) iff

$$\begin{aligned}
\Delta_x + \operatorname{graph}(df_x) = T_xM \times T_xM \\
&\iff \Delta_x \cap \operatorname{graph}(df_x) = 0 \\
&\iff df_x \text{ has no eigenvector of eigenvalue } 1 \\
&\iff \det(df_x - I) \neq 0.
\end{aligned}$$

Definition 16.8. Lefschetz Fixed Point

A fixed point x of f is a *Lefschetz fixed point* if $\det(df_x - I) \neq 0$.

Definition 16.9. Local Lefschetz Number

For a Lefschetz fixed point x of f, the *local Lefschetz number* of f at x, $L_x(f)$, is the orientation number $\in \{\pm 1\}$ of $(x,x) \in \Delta \cap \operatorname{graph}(f)$.

Clearly, for Lefschetz maps,

$$L(f) = \sum_{f(x)=x} L_x(f),$$

the sum being over fixed points.

Proposition 16.10. Formula for Local Lefschetz Number

$$L_x(f) = \operatorname{sign}(\det(df_x - I)).$$

Proof. The proof is simple linear algebra. If $\{v_1, \ldots, v_m\}$ is a positively oriented ordered basis for T_xM, then

$$\begin{aligned}
L_x(f) &= \operatorname{sign}\{\underbrace{(v_1,v_1),\ldots,(v_m,v_m)}_{\text{positive for } T_{(x,x)}\Delta}, \underbrace{(v_1, df_x(v_1)),\ldots,(v_m, df_x(v_m))}_{\text{positive for } T_{(x,x)}\operatorname{graph}(f)}\} \\
&= \operatorname{sign}\{(v_1,v_1),\ldots,(v_m,v_m),(0,(df_x - I)v_1),\ldots,(0,(df_x - I)v_m)\} \\
&= \operatorname{sign}\{(v_1,0),\ldots,(v_m,0),(0,(df_x - I)v_1),\ldots,(0,(df_x - I)v_m)\} \\
&= \operatorname{sign}(\det(df_x - I)).
\end{aligned}$$

□

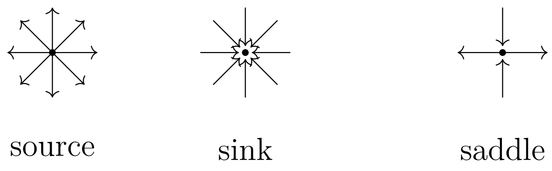
16.1.2 Local Pictures in 2D

Example 16.11. Local Pictures in 2D

Assume that $df_x = \begin{pmatrix} \alpha_1 & 0 \\ 0 & \alpha_2 \end{pmatrix}$ with $\alpha_1, \alpha_2 > 0$.

Case 1: If either $\alpha_1, \alpha_2 > 1$ or $\alpha_1, \alpha_2 < 1$, then $L_x(f) = +1$ (source or sink).

Case 2: If $\alpha_1 < 1 < \alpha_2$, then $L_x(f) = -1$ (saddle).



16.1.3 Euler Characteristic from Local Lefschetz Numbers

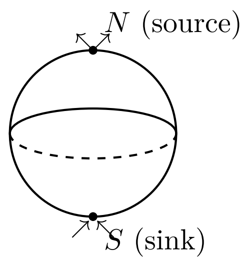
Example 16.12. Euler Characteristic of S^2

Consider the map $f_t\colon S^2 \to S^2$ given by a "downward flow"

$$f_t(x) = \pi\Big(x + \big(0, 0, -\tfrac{t}{2}\big)\Big), \quad 0 < t < 2,$$

where $\pi\colon \mathbb{R}^3 \setminus \{0\} \to S^2$ is the projection. This has a source at the north pole and a sink at the south pole, giving

$$\chi(S^2) = L(f_1) = 1 + 1 = 2.$$



Corollary 16.13. Fixed Points of Maps on S^2

Every map $f\colon S^2 \to S^2$ homotopic to the identity must have a fixed point. In particular, the antipodal map $x \mapsto -x$ is not homotopic to the identity.

More generally, a genus g surface admits a Lefschetz map homotopic to the identity that has 1 source, 1 sink, and $2g$ saddles, giving

$$\chi(\Sigma_g) = 2 - 2g.$$

It is also instructive to see how the fixed points get created and annihilated in pairs when we deform the "height function"; for example, on S^2,

$$\chi(S^2) = 1 + 1 = 1 + 1 + (1 - 1).$$

Chapter 17

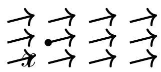
Vector Fields and the Poincaré–Hopf Theorem

17.1 Vector Fields and the Poincaré–Hopf Theorem

Definition 17.1. Vector Field

A *vector field* on a manifold M is a smooth section $v \colon M \to TM$, i.e. a smooth assignment of a tangent vector $v(x) \in T_xM$ at each point $x \in M$.

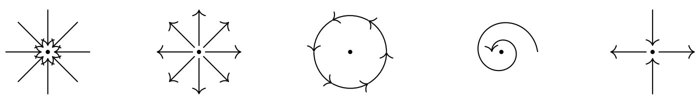
If $v(x) \neq 0$ at some point $x \in M$, then v is nearly constant in magnitude and direction near x.



However, if $v(x) = 0$, v can behave interestingly around such *zeros*.

Example 17.2. Behaviors of Vector Fields Near Zeros

On a surface, some possible behaviors of v around its zeros include: sink, source, circulation, spiral, and saddle.



sink source circulation spiral saddle

We'll see that the zeros of v encode some information on the topology of the manifold M!

17.1.1 Index of a Vector Field

Suppose v is a vector field on $\mathbb{R}^m$ that has an isolated zero at the origin $0 \in \mathbb{R}^m$. Then, for any small enough radius $\varepsilon > 0$, v has no zeros except at the origin inside the sphere S_ε^{m-1} of radius ε around the origin.

Definition 17.3. Index of a Vector Field at a Zero

The *index* of v at 0, $\operatorname{ind}_0(v)$, is the degree of the directional map

$$S_\varepsilon^{m-1} \to S^{m-1}, \quad x \mapsto \frac{v(x)}{|v(x)|}.$$

Example 17.4. Index in Two Dimensions

In the two-dimensional case, $\operatorname{ind}_x(v)$ simply counts the number of times v rotates (in the counterclockwise direction) as we walk counterclockwise around the circle. For example, a sink, source, circulation, and spiral each have index $+1$; a saddle has index -1; and a "double rotation" has index $+2$.

More generally, if x is an isolated zero of a vector field v on M, then we can define $\operatorname{ind}_x(v)$ to be the index of the pushforward vector field $\varphi_*^{-1}(v)$ at 0, where $\varphi\colon U \to M$, $0 \mapsto x$, is a local parametrization.

17.1.2 Poincaré–Hopf Theorem

Theorem 17.5. Poincaré–Hopf Index Theorem

If v is a smooth vector field on a compact, oriented manifold M with only finitely many zeros, then

$$\chi(M) = \sum_{v(x)=0} \operatorname{ind}_x(v).$$

Corollary 17.6. Hairy Ball Theorem

There is no nowhere-vanishing vector field on S^2, since $\chi(S^2) = 2 \neq 0$.
"You can't comb a hairy ball without leaving a bald spot."

Proof of Poincaré–Hopf. The idea is that a vector field determines a flow

$$f_t\colon M \to M$$

that is homotopic to $f_0 = \mathrm{id}$, and tangent to v at time zero. Such a family of maps can be obtained by solving the ODE given by the vector field. Another way is to simply take $f_t(x) = \pi(x + tv(x))$, where π is the projection map from the ε-neighborhood theorem.

For small $t > 0$, fixed points of f_t correspond exactly to zeros of v, so it suffices to prove that $\operatorname{ind}_x(v) = L_x(f_t)$.

This is a local problem, so we may work in $\mathbb{R}^m$. From basic calculus, there is a smooth vector-valued function $r(t,x)$ such that

$$f_t(x) = f_0(x) + tf_0'(x) + t^2 r(t,x) = f_0(x) + tv(x) + t^2 r(t,x),$$

so

$$\frac{f_t(x) - x}{|f_t(x) - x|} = \frac{v(x) + tr(t,x)}{|v(x) + tr(t,x)|}\colon S_\varepsilon^{m-1} \to S^{m-1}$$

is homotopic to $v(x)/|v(x)|$, whose degree is $\operatorname{ind}(v)$.

On the other hand, the local Lefschetz number $L_x(f_t)$ is the degree of the map

$$\partial B_\varepsilon^m \to S^{m-1}, \quad z \mapsto \frac{f_t(z) - z}{|f_t(z) - z|}.$$

For Lefschetz fixed points, simple linear algebra shows that this agrees with the previous definition, since $\deg\Big(z \mapsto \frac{(A-I)z}{|(A-I)z|}\Big) = \operatorname{sign}\det(A - I)$.
Homotopy invariance follows from the boundary theorem. □

Corollary 17.7. Euler Characteristic via Self-Intersection

$\chi(M) = I(M, M)$, where M is considered as the 0-section $M \subset TM$.

Chapter 18

Hopf Degree Theorem

18.1 Hopf Degree Theorem

Let M, N be compact, connected, oriented manifolds of the same dimension. The degree of a map $f \colon M \to N$, which counts the number of preimages (with sign) of a generic point in N, is a homotopy invariant.

In general, the degree is not a complete invariant, but it turns out to be one when $N = S^n$.

Theorem 18.1. Hopf Degree Theorem

Let M be a compact, connected, oriented m-manifold. Two maps $f_0, f_1 \colon M \to S^m$ are homotopic if and only if

$$\deg f_0 = \deg f_1.$$

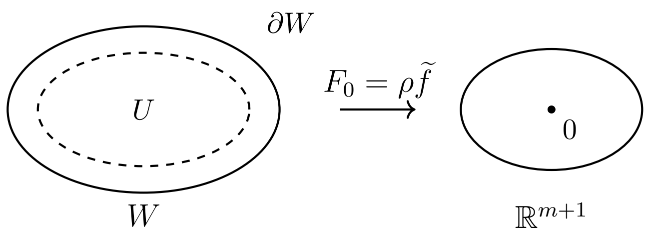
Proof. We already know ($\Rightarrow$).
For ($\Leftarrow$), set $W = M \times I$, and let $f \colon \partial W \to S^m$ be f_0 on $M \times \{0\}$ and f_1 on $M \times \{1\}$, so that $\deg f = \deg f_1 - \deg f_0 = 0$. The claim then follows from the more general extension theorem below. □

18.1.1 Extension Theorem

Theorem 18.2. Extension Theorem

Let W be a compact, connected, oriented $(m+1)$-manifold with boundary, and let $f \colon \partial W \to S^m$ be a smooth map. Then f extends to a map $F \colon W \to S^m$ with $\partial F = f$ if and only if $\deg f = 0$.

Proof. We already know ($\Rightarrow$), so let's show ($\Leftarrow$).
Firstly, any smooth map $f \colon \partial W \to \mathbb{R}^{m+1}$ (without any degree constraint) can be extended to all of W. Let U be a collar neighborhood of ∂W and $\tilde{f} \colon U \xrightarrow{\pi} \partial W \xrightarrow{f} \mathbb{R}^{m+1}$ the projection composed with f. Choose a smooth bump $\rho \colon W \to \mathbb{R}$ with $\rho|_{\partial W} \equiv 1$ and $\rho \equiv 0$ outside a compact subset of U. Then $F_0 := \rho \tilde{f}$ extends f.



After some homotopy in the interior, we may assume that $0 \in \mathbb{R}^{m+1}$ is a regular value of F_0, so $F_0^{-1}(0)$ is a finite set of points in $\mathrm{Int}(W)$. We will need:

Lemma 18.3. Isotopy Lemma

Given any two points x, y in a connected manifold M, there exists a diffeomorphism $h\colon M \to M$ such that $h(x) = y$ which is isotopic to the identity (i.e. each h_t is a diffeomorphism). The isotopy may be taken to be compactly supported.

Using the isotopy lemma, we may place the finite set $F_0^{-1}(0)$ inside a ball $B \subset \mathrm{Int}(W)$. By the assumption $\deg f = 0$, the directional map $\frac{F_0}{|F_0|}\Big|_{\partial B}\colon \partial B \to S^m$ has degree 0, i.e. $F_0|_{\partial B}$ has winding number 0 around $0 \in \mathbb{R}^{m+1}$. To show that $\frac{F_0}{|F_0|}\Big|_{W\setminus\mathrm{Int}(B)}$ extends to all of W, it then suffices to prove the following special case of the Hopf degree theorem. $\square$

18.1.2 Hopf Theorem for Spheres

Theorem 18.4. Hopf Theorem for Spheres

Any smooth map $f\colon S^m \to S^m$ of degree 0 is homotopic to a constant map.

Proof. We proceed by induction on m.
The case $m = 1$ is an exercise (lift the map to a map $\mathbb{R} \to \mathbb{R}$).
Now assume the claim is true for $m - 1$, and we'll extend it to m. Let $p \in S^m$ be a regular value. Using the isotopy lemma, we may assume that the preimage points $f^{-1}(p)$ are contained inside a ball $B \subset S^m$ such that $q \notin f(B)$ for some $q \in S^m$.
Then we have

$$S^{m-1} \cong \partial B \xrightarrow{f|_{\partial B}} S^m \setminus \{q\} \cong \mathbb{R}^m,$$

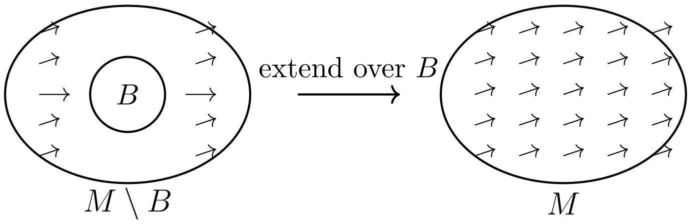
where the last identification is some diffeomorphism sending p to $0 \in \mathbb{R}^m$. This composition has winding number 0 around $0 \in \mathbb{R}^m$, since $\deg f = 0$.
By the induction hypothesis, $f|_{\partial B}$ is homotopic to a constant map, without crossing p.
Therefore $f|_{S^m\setminus B}\colon S^m \setminus B \to S^m \setminus \{p\}$ extends to the whole S^m, and thus $f\colon S^m \to S^m$ is homotopic to a constant map. $\square$

18.1.3 Application: Nowhere-Vanishing Vector Fields

Theorem 18.5. Existence of Nowhere-Vanishing Vector Fields

A compact, connected, oriented manifold M possesses a nowhere-vanishing vector field if and only if $\chi(M) = 0$.

Proof. From Poincaré–Hopf, we already know ($\Rightarrow$).
For the converse ($\Leftarrow$), take a generic vector field v on M with finitely many zeros. Using the isotopy lemma, we may place all the zeros inside some ball $B \subset M$, and since $\chi(M) = 0$, the sum of the indices of the zeros equals 0.
It follows that the map $v\colon S^{m-1} \cong \partial B \to \mathbb{R}^m \setminus \{0\}$ has winding number 0 around $0 \in \mathbb{R}^m$, and by the Hopf degree theorem, it is homotopic to a constant map. Using this homotopy, we can extend $v|_{M\setminus B}$ to a nowhere-vanishing vector field on all of M. $\square$



Chapter 19

Triangulations and Cobordisms

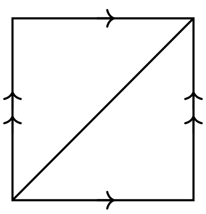
19.1 Euler Characteristics and Triangulations

Definition 19.1. Triangulation

A *triangulation* of a manifold is a subdivision into *simplices*, the higher-dimensional generalization of triangles.

Example 19.2. Triangulation of T^2

The torus T^2, viewed as a square with opposite sides identified, can be triangulated by adding a diagonal, yielding 1 vertex, 3 edges, and 2 triangular faces.

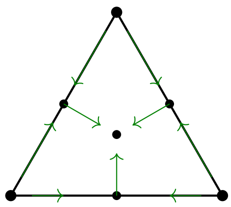


Theorem 19.3. Euler Characteristic via Triangulation

If M is an m-dimensional triangulated manifold, then

$$\chi(M) = \sum_{j=0}^{m} (-1)^j \cdot \#(j\text{-dimensional faces}).$$

Proof. Given a triangulation, we can construct a vector field v on M with exactly one zero on each face, where the zero on a j-dimensional face has index $(-1)^j$:



The theorem then follows from Poincaré–Hopf. □

19.2 Cobordisms (extra topic)

Much of our discussion so far can be generalized to intersection theory even when the dimensions are not complementary.

Suppose $M_1, M_2 \subset N$ are compact manifolds (of not necessarily complementary dimension). Then, after some homotopy to make the intersection transverse, $M_1 \cap M_2$ is a manifold of dimension $m_1 + m_2 - n$.

Under a homotopy of M_1 (or M_2), $M_1 \cap M_2$ may change shape, but it remains in the same *cobordism class*: tracking the intersection across the homotopy traces out a manifold inside $N \times I$ whose boundary is the two intersections at times 0 and 1.

Definition 19.4. Cobordant Manifolds

Two manifolds M, M' of the same dimension are *cobordant* if there is a compact manifold W with boundary $\partial W \cong M \sqcup M'$.

Being cobordant is an equivalence relation, and we call the equivalence classes *cobordism classes*. They form an abelian group under disjoint union.

Definition 19.5. Cobordism Groups

Let Ω_n^O be the *unoriented cobordism group* of n-manifolds, and let Ω_n^{SO} be the *oriented cobordism group* of oriented n-manifolds (and oriented cobordisms).

The Cartesian product defines a multiplication, so

$$\Omega_*^O := \bigoplus_{n\geq 0} \Omega_n^O \quad \text{and} \quad \Omega_*^{SO} := \bigoplus_{n\geq 0} \Omega_n^{SO}$$

are graded algebras.

Example 19.6. Zero-Dimensional Cobordism Groups

$$\Omega_0^O \cong \mathbb{Z}/2 \quad \text{(generated by a point)}, \qquad \Omega_0^{SO} \cong \mathbb{Z} \quad \text{(generated by a signed point)}.$$

These are exactly where our mod 2 and oriented intersection theory took values!

19.2.1 Thom's Theorem

This is out of the scope for this course, but René Thom proved in 1954:

Theorem 19.7. Thom's Theorem (Unoriented Cobordism)

$$\Omega_*^O \cong \mathbb{F}_2[x_i \mid i \geq 1,\ i \neq 2^j - 1],$$

a polynomial algebra with one generator x_i in each dimension $i \neq 2^j - 1$. For even i, we may take $x_i = [\mathbb{R}P^i]$.

Explicitly, in low dimensions:

$$\Omega_0^O \cong \mathbb{Z}/2, \quad \Omega_1^O \cong 0, \quad \Omega_2^O \cong \mathbb{Z}/2, \quad \Omega_3^O \cong 0, \quad \Omega_4^O \cong \mathbb{Z}/2 \oplus \mathbb{Z}/2, \quad \Omega_5^O \cong \mathbb{Z}/2, \ldots$$

For oriented cobordisms, it is known that, modulo torsion,

$$\Omega_*^{SO} \otimes \mathbb{Q} \cong \mathbb{Q}[y_{4i} \mid i \geq 1], \quad \text{where } y_{4i} = [\mathbb{C}P^{2i}],$$

and

$$\Omega_0^{SO} \cong \mathbb{Z}, \quad \Omega_1^{SO} \cong 0, \quad \Omega_2^{SO} \cong 0, \quad \Omega_3^{SO} \cong 0, \quad \Omega_4^{SO} \cong \mathbb{Z}, \quad \Omega_5^{SO} \cong \mathbb{Z}/2, \dots$$

Chapter 20

Differential Forms

20.1 Differential Forms

Recall that a vector field v on M is a smoothly varying field of tangent vectors $v_x \in T_xM$.

Definition 20.1. 1-Form

A *1-form* α on M is a smoothly varying field of cotangent vectors

$$\alpha_x \in T_x^*M := (T_xM)^*.$$

If v is a vector field and α is a 1-form, then

$$\alpha(v)\colon M \to \mathbb{R}, \qquad x \mapsto \alpha_x(v_x),$$

is a smooth function.

Example 20.2. Differential of a Function

If $f\colon M \to \mathbb{R}$ is a smooth function, then $df_x\colon T_xM \to \mathbb{R}$ is a linear map at each x, so df is a 1-form on M. For any vector field v on M,

$$df(v) =: vf$$

is a smooth function on M, given by the directional derivative of f along v_x at each $x \in M$.

Definition 20.3. p-Form

A *p-form* ω on M is a smoothly varying field of alternating p-tensors

$$\omega_x \in \Lambda^p(T_x^*M).$$

(A 0-form is just a smooth function.)

20.1.1 Forms on Euclidean Space

Example 20.4. Coordinate Forms

In $\mathbb{R}^m$, the coordinate functions $x_1, \dots, x_m$ yield 1-forms $dx_1, \dots, dx_m$. For each strictly increasing sequence $I = (i_1, \dots, i_p)$ with $1 \le i_1 < \cdots < i_p \le m$,

$$dx_I := dx_{i_1} \wedge \cdots \wedge dx_{i_p}$$

is a p-form on $\mathbb{R}^m$.

In fact, every p-form on an open subset $U \subset \mathbb{R}^m$ can be uniquely expressed as a sum

$$\omega = \sum_I f_I \, dx_I$$

over increasing index sequences I, the f_I being functions on U.

20.1.2 Properties of Differential Forms

- **(Linearity)** If ω_1 and ω_2 are p-forms, then their pointwise sum $\omega_1 + \omega_2$ is also a p-form.
- **(Wedge product)** If ω is a p-form and θ is a q-form, then $\omega \wedge \theta = (-1)^{pq} \, \theta \wedge \omega$ is a $(p+q)$-form.
- **(Pullback)** If $f\colon M \to N$ is a smooth map and ω is a p-form on N, then the *pullback* $f^*\omega$ of ω by f, defined by

  $$(f^*\omega)_x = (df_x)^* \omega_{f(x)},$$

  is a p-form on M. Equivalently, $(f^*\omega)_x$ is the composition

  $$\Lambda^p(T_x M) \xrightarrow{df_x} \Lambda^p(T_{f(x)} N) \xrightarrow{\omega_{f(x)}} \mathbb{R}.$$

  The pullback satisfies

  $$f^*(\omega_1 + \omega_2) = f^*\omega_1 + f^*\omega_2, \qquad f^*(\omega \wedge \theta) = f^*\omega \wedge f^*\theta, \qquad (f \circ g)^*\omega = g^* f^* \omega.$$

20.1.3 Pullback on Euclidean Space

Example 20.5. Pullback in Coordinates

Let $f\colon V \to U$ be a smooth map between open subsets $U \subset \mathbb{R}^m$, $V \subset \mathbb{R}^n$, with coordinate functions $x_1, \dots, x_m$ on $\mathbb{R}^m$ and $y_1, \dots, y_n$ on $\mathbb{R}^n$. Writing $f_i := x_i \circ f$,

$$f^* dx_i = \sum_{j=1}^n \frac{\partial f_i}{\partial y_j} \, dy_j = df_i.$$

For an arbitrary p-form $\omega = \sum_I a_I \, dx_I$ on U,

$$f^*\omega = \sum_I (f^* a_I)(f^* dx_I) = \sum_I (a_I \circ f) \, df_I,$$

where $df_I = df_{i_1} \wedge \cdots \wedge df_{i_p}$ for $I = (i_1, \dots, i_p)$.

Example 20.6. Volume Form Under a Diffeomorphism

In the special case when $f\colon V \to U$ is a diffeomorphism of two open subsets of $\mathbb{R}^m$,

$$f^*(\underbrace{dx_1 \wedge \cdots \wedge dx_m}_{\text{“volume form”}}) = df_1 \wedge \cdots \wedge df_m = \det(df)\, dy_1 \wedge \cdots \wedge dy_m.$$

20.1.4 Smoothness of Forms

We can use the pullback to define precisely what we mean by “smoothness” for a form ω.

A form ω on an open set $U \subset \mathbb{R}^m$ is *smooth* if each coefficient function a_I in its expansion $\omega = \sum_I a_I\, dx_I$ is smooth. Note that if ω is smooth and $f\colon V \to U$ is a smooth map, then $f^*\omega$ is smooth as well.

More generally, a form ω on M is *smooth* if, for every local parametrization $\varphi\colon U \to M$, the pullback $\varphi^*\omega$ is smooth on U.

Chapter 21

Integration on Manifolds

21.1 Integration on Manifolds

21.1.1 Change of Variables in $\mathbb{R}^m$

Recall from vector calculus that, if $f\colon V \to U$ is a diffeomorphism of open sets in $\mathbb{R}^m$ and a is an integrable function on U, then

$$\int_U a\, dx_1 \cdots dx_m = \int_V (a \circ f)\, |\det(df)|\, dy_1 \cdots dy_m.$$

Notice the similarity with the pullback formula:

$$f^*(a\, dx_1 \wedge \cdots \wedge dx_m) = (a \circ f)\, \det(df)\, dy_1 \wedge \cdots \wedge dy_m.$$

The two differ only in the sign $\det(df)$ vs. $|\det(df)|$. It follows that:

Theorem 21.1. Change of Variables in $\mathbb{R}^m$

If $f\colon V \to U$ is an orientation-preserving diffeomorphism of open sets in $\mathbb{R}^m$ (or H^m) and ω is an integrable m-form on U, then

$$\int_U \omega = \int_V f^*\omega.$$

If f reverses the orientation, then $\int_U \omega = -\int_V f^*\omega$.

21.1.2 Integral of an m-Form over a Manifold

Definition 21.2. Integral of an m-Form

Let M be an oriented m-manifold and let ω be a smooth, compactly supported m-form on M. Define the *integral of ω over M* to be

$$\int_M \omega := \sum_i \int_{U_i} \rho_i \omega,$$

a finite sum, where $\{\rho_i\}$ is a partition of unity subordinate to parametrizable open subsets and each integral is defined in some local parametrization.

The change of variables formula ensures that the integral is well-defined, independent of the choice of parametrizations. In particular:

Theorem 21.3. Naturality of the Integral

If $f\colon N \to M$ is an orientation-preserving diffeomorphism, then

$$\int_M \omega = \int_N f^*\omega$$

for any compactly supported smooth m-form ω (where $m = \dim M = \dim N$).

That is, the integral transforms naturally!

21.1.3 Integration over Submanifolds

Just like we can integrate an m-form over M, we can integrate any p-form ω over an oriented p-dimensional submanifold $i\colon P \hookrightarrow M$ by restricting ω to P:

$$\int_P \omega := \int_P i^*\omega.$$

Example 21.4. Line Integral of a 1-Form on $\mathbb{R}^3$

Suppose $\omega = f_1\,dx_1 + f_2\,dx_2 + f_3\,dx_3$ is a smooth 1-form on $\mathbb{R}^3$, and $\gamma\colon I \to \mathbb{R}^3$ is a diffeomorphism of $I = [0,1]$ onto a curve $C = \gamma(I)$. Then

$$\int_C \omega = \int_I \gamma^*\omega = \int_0^1 \sum_{i=1}^3 f_i(\gamma(t))\,d\gamma_i = \int_0^1 \sum_{i=1}^3 f_i(\gamma(t))\,\frac{d\gamma_i}{dt}(t)\,dt.$$

Example 21.5. Surface Integral of a 2-Form on $\mathbb{R}^3$

Suppose

$$\omega = f_1\,dx_2 \wedge dx_3 + f_2\,dx_3 \wedge dx_1 + f_3\,dx_1 \wedge dx_2$$

is a compactly supported smooth 2-form on $\mathbb{R}^3$. Let $S \subset \mathbb{R}^3$ be a surface, which we assume for simplicity to be the graph of a function $G\colon \mathbb{R}^2 \to \mathbb{R}$, i.e.

$$S = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid x_3 = G(x_1, x_2)\}.$$

We use the parametrization

$$h\colon \mathbb{R}^2 \to S, \qquad (x_1, x_2) \mapsto (x_1, x_2, G(x_1, x_2)).$$

Then $h^*dx_3 = dG = \frac{\partial G}{\partial x_1}dx_1 + \frac{\partial G}{\partial x_2}dx_2$, so

$$\begin{aligned}
\int_S \omega &= \int_{\mathbb{R}^2} h^*\omega \\
&= \int_{\mathbb{R}^2} (f_1 \circ h)\,dx_2 \wedge dG + (f_2 \circ h)\,dG \wedge dx_1 + (f_3 \circ h)\,dx_1 \wedge dx_2 \\
&= \int_{\mathbb{R}^2} \Big((f_1 \circ h)\Big(-\tfrac{\partial G}{\partial x_1}\Big) + (f_2 \circ h)\Big(-\tfrac{\partial G}{\partial x_2}\Big) + (f_3 \circ h)\Big)\,dx_1 \wedge dx_2.
\end{aligned}$$

Note that

$$\vec{n} := \Big(-\tfrac{\partial G}{\partial x_1}, -\tfrac{\partial G}{\partial x_2}, 1\Big)$$

is normal to $T_xS = \text{Span}\{(1, 0, \frac{\partial G}{\partial x_1}), (0, 1, \frac{\partial G}{\partial x_2})\}$ at every $x \in S$. It follows that we can rewrite the integral as

$$\int_S \omega = \int_{\mathbb{R}^2} (\vec{F} \cdot \vec{u})\, dA,$$

where $\vec{F} = (f_1, f_2, f_3)$, $\vec{u} = \vec{n}/|\vec{n}|$, and $dA = |\vec{n}|\, dx_1 \wedge dx_2$ — a form familiar from vector calculus.

Chapter 22

Exterior Derivative and de Rham Cohomology

22.1 Exterior Derivative

Let $\Omega^p(M)$ denote the set of all smooth p-forms on M. These are $\mathbb{R}$-vector spaces, and in fact $C^\infty(M)$-modules, where $C^\infty(M) = \Omega^0(M)$.

Recall the linear map

$$\Omega^0(M) \xrightarrow{d} \Omega^1(M), \qquad f \mapsto df.$$

This can be extended to arbitrary p-forms. On an open subset $U \subset \mathbb{R}^m$, define the *exterior derivative*

$$\Omega^p(U) \xrightarrow{d} \Omega^{p+1}(U), \qquad \omega = \sum_I a_I \, dx_I \mapsto d\omega = \sum_I da_I \wedge dx_I.$$

Theorem 22.1. Properties of the Exterior Derivative

The exterior derivative d on smooth forms on $U \subset \mathbb{R}^m$ satisfies:

(a) **(Linearity)** $d(\omega_1 + \omega_2) = d\omega_1 + d\omega_2$.

(b) **(Multiplication law)** $d(\omega \wedge \theta) = d\omega \wedge \theta + (-1)^p \omega \wedge d\theta$, if ω is a p-form.

(c) **(Cocycle condition)** $d(d\omega) = 0$.

Furthermore, this is the unique operator with these properties that agrees with the previous definition of df for smooth functions f.

Proof. Linearity is obvious, and the multiplication law is an easy computation. The cocycle condition is also a straightforward computation:

$$\omega = \sum_I a_I \, dx_I \implies d\omega = \sum_I da_I \wedge dx_I = \sum_I \left(\sum_i \frac{\partial a_I}{\partial x_i} \, dx_i \right) \wedge dx_I,$$

so

$$d(d\omega) = \sum_I \left(\sum_i \sum_j \frac{\partial^2 a_I}{\partial x_i \partial x_j} \, dx_j \wedge dx_i \right) \wedge dx_I = 0,$$

since the (i,j)-term cancels with the (j,i)-term.

For uniqueness, if D were another operator satisfying (a), (b), (c) and $Df = df$, then by (a) and (b),

$$\begin{aligned} D\omega = D\left(\sum_I a_I\, dx_I\right) &= \sum_I \left(Da_I \wedge dx_I + a_I\, D(dx_I)\right) \\ &= \sum_I \left(da_I \wedge dx_I + a_I \sum_{1\le j\le p} (-1)^{j-1}\, dx_{i_1} \wedge \cdots \wedge D\, dx_{i_j} \wedge \cdots \wedge dx_{i_p}\right) \\ &= \sum_I da_I \wedge dx_I = d\omega, \end{aligned}$$

where the last equality uses (c) applied to $D\, dx_{i_j} = D(Dx_{i_j}) = 0$. ☐

Corollary 22.2. Diffeomorphism Invariance

If $g\colon V \to U$ is a diffeomorphism of open subsets of $\mathbb{R}^m$, then for every form ω on U,

$$d(g^*\omega) = g^*(d\omega).$$

Proof. The operator $D := (g^{-1})^* \circ d \circ g^*$ satisfies (a), (b), (c) and $D = d$ on functions. By uniqueness, $D = d$, i.e. $d \circ g^* = g^* \circ d$. ☐

This corollary allows us to define

$$\Omega^p(M) \xrightarrow{d} \Omega^{p+1}(M)$$

locally, using local parametrizations: the corollary says it is independent of the choice of local parametrizations. The properties of exterior derivatives on Euclidean spaces carry over to arbitrary manifolds with boundary. Moreover, d commutes with arbitrary pullback:

Theorem 22.3. d Commutes with Pullback

If $g\colon N \to M$ is any smooth map of manifolds, then for any form ω on M,

$$d(g^*\omega) = g^*(d\omega).$$

Proof. This holds for any 0-form (exercise) and also when $\omega = df$, since

$$d(g^*df) = d(d\, g^*f) = 0 = g^*(d\, df).$$

If the theorem holds for ω and θ, then it holds for $\omega \wedge \theta$:

$$\begin{aligned} d(g^*(\omega \wedge \theta)) = d(g^*\omega \wedge g^*\theta) &= d(g^*\omega) \wedge g^*\theta + (-1)^p\, g^*\omega \wedge d(g^*\theta) \\ &= g^*d\omega \wedge g^*\theta + (-1)^p\, g^*\omega \wedge g^*d\theta = g^*\, d(\omega \wedge \theta). \end{aligned}$$

Every form on M is locally expressible as $\sum_I a_I\, dx_I$, and since the theorem is local, it holds for every form. ☐

Example 22.4. Explicit Calculation of d on $\mathbb{R}^3$

The de Rham sequence on $\mathbb{R}^3$,

$$\Omega^0(\mathbb{R}^3) \xrightarrow{d} \Omega^1(\mathbb{R}^3) \xrightarrow{d} \Omega^2(\mathbb{R}^3) \xrightarrow{d} \Omega^3(\mathbb{R}^3) \xrightarrow{d} 0,$$

recovers the classical vector calculus operators, using the shorthand $\partial_i := \partial/\partial x_i$.
Gradient. $f \mapsto \partial_1 f\, dx_1 + \partial_2 f\, dx_2 + \partial_3 f\, dx_3$.
Curl. $f_1\, dx_1 + f_2\, dx_2 + f_3\, dx_3 \mapsto (\partial_2 f_3 - \partial_3 f_2)\, dx_2 \wedge dx_3 + \text{cyc.}$
Divergence. $f_1\, dx_2 \wedge dx_3 + f_2\, dx_3 \wedge dx_1 + f_3\, dx_1 \wedge dx_2 \mapsto (\partial_1 f_1 + \partial_2 f_2 + \partial_3 f_3)\, dx_1 \wedge dx_2 \wedge dx_3$.

22.2 de Rham Cohomology

On any smooth manifold M, we have a chain complex

$$0 \to \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^m(M) \to 0,$$

called the *de Rham complex*.

Definition 22.5. Closed and Exact Forms

A p-form ω is *closed* if $d\omega = 0$ (i.e. $\omega \in \ker d$), and *exact* if $\omega = d\theta$ for some $(p-1)$-form θ (i.e. $\omega \in \operatorname{im} d$).

Since $d^2 = 0$, every exact form is closed, but the converse is not always true.

Definition 22.6. de Rham Cohomology Group

The *p-th de Rham cohomology group* of M is the $\mathbb{R}$-vector space of closed p-forms modulo exact p-forms,

$$H^p(M) := \frac{\ker\Big(\Omega^p(M) \xrightarrow{d} \Omega^{p+1}(M)\Big)}{\operatorname{im}\Big(\Omega^{p-1}(M) \xrightarrow{d} \Omega^p(M)\Big)}.$$

Remark 22.7. Functoriality

Since d commutes with pullbacks, any smooth map $f\colon N \to M$ induces a linear map $f^*\colon H^p(M) \to H^p(N)$ on de Rham cohomology groups.

Example 22.8. H^0

A 0-form (i.e. a function) is closed iff it is constant on each component, so

$$\dim H^0(M) = \#\,\text{connected components of } M.$$

Remark 22.9. Cohomology of $\mathbb{R}^m$ and S^m

$$H^p(\mathbb{R}^m) = \begin{cases} \mathbb{R} & \text{if } p = 0, \\ 0 & \text{if } p > 0, \end{cases} \qquad H^p(S^m) = \begin{cases} \mathbb{R} & \text{if } p = 0 \text{ or } m, \\ 0 & \text{otherwise} \end{cases} \qquad (m > 0).$$

Chapter 23

Stokes Theorem

23.1 Stokes Theorem

Stokes theorem is a remarkable theorem relating d with ∂, which generalizes the fundamental theorem of calculus.

Theorem 23.1. Stokes Theorem

Let M be a compact, oriented m-manifold with boundary. Then, for any smooth $(m-1)$-form ω on M,

$$\int_{\partial M} \omega = \int_M d\omega.$$

Before getting into the proof, let's first appreciate this theorem for a moment.

Remark 23.2. Special Cases of Stokes

- When $M = [a,b] \subset \mathbb{R}$ and $\omega\colon [a,b] \to \mathbb{R}$ is a function, this is just the fundamental theorem of calculus.
- When M is boundaryless, the integral of any exact form on a boundaryless manifold is 0.
- If ω is a closed $(m-1)$-form on M, then $\int_{\partial M} \omega = 0$.
- The theorem is compatible with $\partial^2 = 0$ and $d^2 = 0$:

$$0 = \int_{\partial^2 M} \omega = \int_{\partial M} d\omega = \int_M d^2\omega = 0.$$

Proof of Stokes. Since both sides of the equation are linear in ω, we may assume that ω has compact support contained in a parametrizable open subset U. That is, we can work locally in an open subset of $\mathbb{R}^m$ or H^m.
Any $(m-1)$-form in m-space may be written as

$$\nu = \sum_{i=1}^{m} (-1)^{i-1} f_i \, dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_m,$$

where $\widehat{dx_i}$ means dx_i is omitted. So

$$d\nu = \left(\sum_{i=1}^{m} \frac{\partial f_i}{\partial x_i}\right) dx_1 \wedge \cdots \wedge dx_m.$$

Case 1: U open in $\mathbb{R}^m$. If ν has compact support in U, then by Fubini and the fundamental theorem of calculus,

$$\int_U d\nu = \sum_i \int_{\mathbb{R}^m} \frac{\partial f_i}{\partial x_i}\, dx_1 \cdots dx_m = \sum_i \int_{\mathbb{R}^{m-1}} \left(\int_{-\infty}^{\infty} \frac{\partial f_i}{\partial x_i}\, dx_i\right) dx_1 \cdots \widehat{dx_i} \cdots dx_m = 0,$$

since each f_i has compact support. And $\partial U = \emptyset$, so $\int_{\partial U} \nu = 0$ as well.
Case 2: U open in H^m. By Fubini, only the $i = m$ term survives integration over H^m:

$$\begin{aligned}\int_U d\nu &= \sum_i \int_{H^m} \frac{\partial f_i}{\partial x_i}\, dx_1 \cdots dx_m = \int_{\mathbb{R}^{m-1}} \left(\int_0^{\infty} \frac{\partial f_m}{\partial x_m}\, dx_m\right) dx_1 \cdots dx_{m-1} \\ &= \int_{\mathbb{R}^{m-1}} -f_m(x_1, \dots, x_{m-1}, 0)\, dx_1 \cdots dx_{m-1} =: \circledast,\end{aligned}$$

by the fundamental theorem of calculus.
On the other hand, since $x_m = 0$ on ∂H^m, we have $dx_m = 0$ on ∂H^m, so only the $i = m$ term in ν survives, giving

$$\begin{aligned}\int_{\partial H^m} \nu &= \int_{\partial H^m} (-1)^{m-1} f_m(x_1, \dots, x_{m-1}, 0)\, dx_1 \wedge \cdots \wedge dx_{m-1} \\ &= (-1)^m \int_{\mathbb{R}^{m-1}} (-1)^{m-1} f_m(x_1, \dots, x_{m-1}, 0)\, dx_1 \cdots dx_{m-1} = \circledast,\end{aligned}$$

where the leading $(-1)^m$ records the change of orientation under the diffeomorphism $\partial H^m \to \mathbb{R}^{m-1}$, $(x_1, \dots, x_{m-1}, 0) \mapsto (x_1, \dots, x_{m-1})$, given by the sign of the ordered basis $\{-e_m, e_1, \dots, e_{m-1}\}$. □

23.1.1 Consequences

Corollary 23.3. Integration Defines a Cohomology Functional

If P is a p-dimensional compact, oriented submanifold of M, then integration over P defines a linear functional

$$\int_P : H^p(M) \to \mathbb{R}.$$

Moreover, if $P = \partial W$ for some compact, oriented $(p+1)$-dimensional submanifold with boundary in M, then $\int_P = 0$.

Corollary 23.4. Vanishing under Bounding Extension

If $f : M \to N$ is a smooth map of compact, oriented m-manifolds which extends to all of a compact, oriented $(m+1)$-manifold W with $\partial W = M$, then

$$\int_M f^*\omega = 0$$

for any m-form ω on N.

Proof. Let $F\colon W \to N$ extend f. Then by Stokes,

$$\int_M f^*\omega = \int_{\partial W} F^*\omega = \int_W F^* d\omega = 0,$$

since ω is a top-degree form, so $d\omega = 0$. $\square$

Corollary 23.5. Homotopy Invariance of the Integral

If $f_0, f_1\colon M \to N$ are homotopic maps of compact, oriented m-manifolds, then for every m-form ω on N,

$$\int_M f_0^*\omega = \int_M f_1^*\omega.$$

Proof. Apply the previous corollary to the homotopy $F\colon I \times M \to N$. $\square$

Remark 23.6. Degree Formula

While we won't have time to cover it, the following *degree formula* holds:

$$\int_M f^*\omega = \deg(f) \int_N \omega.$$